

# Conformational dynamics and asymmetry in multimodal inhibition of membrane-bound pyrophosphatases

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## eLife Assessment

This **important** study uncovers the mechanism of inhibition of a membrane pyrophosphatase by non-hydrolyzable phosphonate substrate analogs. **Convincing** crystallography, EPR spectroscopy, and functional measurements support the presence of a distinct conformational equilibrium of TmPPase in solution, and further supports the notion of asymmetric inhibitor binding at the active site, while maintaining a symmetric conformation at the periplasmic interface.

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## Abstract

Membrane-bound pyrophosphatases (mPPases) are homodimeric proteins that hydrolyse pyrophosphate and pump  $H^+/Na^+$  across membranes. They are crucial for the virulence of protist pathogens, making them attractive drug targets. In this study, we investigate the inhibitory effects of seven distinct bisphosphonates against *Thermotoga maritima* mPPase to explore their mode of action and assist in future small molecule inhibitor development. We solved two structures of mPPase bound to the inhibitors in the enzyme active sites and probed the conformational dynamics of mPPase under multiple inhibitors and functionally relevant conditions by double electron-electron resonance (DEER) spectroscopy. We found that mPPase adopts distinct conformational equilibria in solution in the presence of different inhibitors, including states consistent with asymmetric binding in the active site (closed-open), but a symmetric apo-like conformation on the periplasmic side (open-open). Combined with solid-supported membrane-based electrophysiology recordings, this revealed that

during catalysis, one monomer of the dimer remains open, and  $\text{Na}^+$  can only be pumped in a closed state. These results further support symmetry-breaking across the membrane, consistent with half-of-the-sites-reactivity.

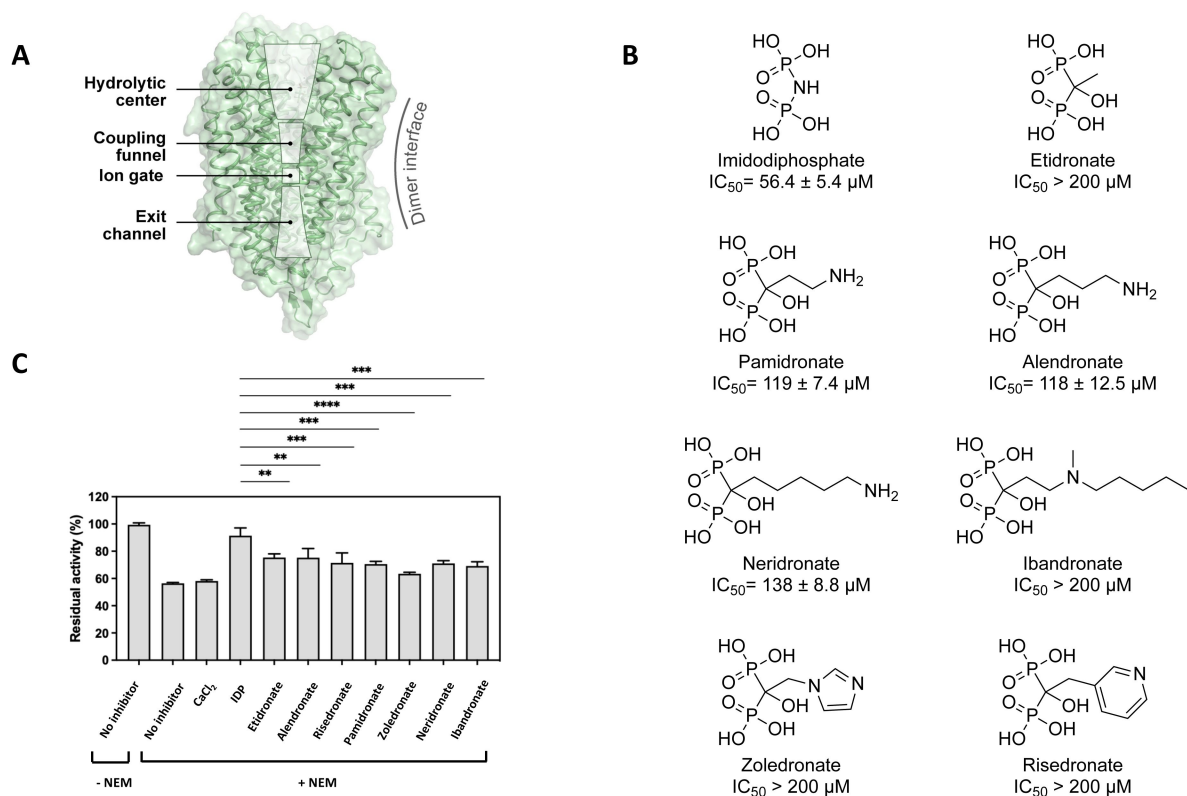
## Introduction

Membrane-bound pyrophosphatases (mPPases) facilitate the transport of protons and/or sodium ions across membranes while catalysing the breakdown of pyrophosphate ( $\text{PP}_i$ ), a by-product generated in various cellular synthetic reactions - into inorganic phosphate ( $\text{P}_i$ ). These enzymes are found in plants, certain species of bacteria, protist parasites, and archaea, but are absent from multicellular animals<sup>1,2,3,4</sup>. Within these organisms, mPPases are essential for cell survival under diverse stress conditions such as osmotic stress, mineral deficiency, and extreme temperature<sup>6</sup>. Based on their potassium dependency, mPPases are divided into two families:  $\text{K}^+$ -dependent and  $\text{K}^+$ -independent. While  $\text{K}^+$ -independent mPPases all transport  $\text{H}^+$ ,  $\text{K}^+$ -dependent mPPases can transport  $\text{H}^+$ ,  $\text{Na}^+$ , or both<sup>4</sup>.

Currently, mPPase structures have been reported from three different organisms: *Vigna radiata* (VrPPase), *Thermotoga maritima* (TmPPase), and, most recently, a structure from *Pyrobaculum aerophilum* (PaPPase) in complex with imidodiphosphate (IDP)<sup>7</sup>. For TmPPase, several different structural states have been determined, including the resting state (TmPPase:Ca:Mg)<sup>8</sup>, with two phosphates bound (TmPPase:2P)<sup>8</sup>, IDP bound (TmPPase:IDP)<sup>9</sup>, IDP and *N*-[(2-amino-6-benzothiazolyl)methyl]-1*H*-indole-2-carboxamide (ATC) bound (TmPPase:IDP:ATC)<sup>10</sup>, phosphate analogue (WO)-bound (TmPPase:WO)<sup>9</sup>, and time-resolved X-ray diffraction structures (with and without substrate/product bound) showing structural asymmetry<sup>7</sup>. Similarly, VrPPase has been solved in multiple states, including IDP-bound (VrPPase:IDP)<sup>11</sup>, single phosphate-bound (VrPPase:P)<sup>9</sup>, two phosphates bound (VrPPase:2P), and different mutations at the hydrophobic gate<sup>12</sup>. These structures show that mPPases are homodimeric enzymes, with each monomer consisting of 16 and, as found in sequence databases, occasionally 17 transmembrane helices (TMHs), organised into two concentric rings: the inner ring (TMH5-6, 11-12, and 15-16) and the outer ring (TMH1-4, 7-10, and 13-14). Each monomer consists of four regions: a hydrolytic centre, a coupling funnel, an ion gate, and an exit channel<sup>13</sup> (**Fig. 1A**). To simplify residue comparison between mPPases, we employ the residue numbering scheme  $\text{XZ}^{\text{Y,Z}}$  (superscripts refer to Ballesteros–Weinstein numbering<sup>14</sup>), where X represents the amino acid, Z denotes the amino acid position in TmPPase, Y indicates the helix number, and Z specifies the offset of amino acid positions within the centrally conserved residues of the helix<sup>13</sup>.

mPPases are a promising drug target for treating diseases caused by parasitic protists, such as malaria and leishmaniasis<sup>10,15,16,17</sup>. Among the currently available compounds, ATC demonstrates the most effective inhibitory activity against TmPPase<sup>10</sup>. ATC is bound to a region near the enzyme exit channel of one subunit, which induces structural asymmetry in the mPPase dimer<sup>10</sup>. Functional asymmetry in  $\text{K}^+$ -dependent mPPases has also previously been shown by Artukka, et al.<sup>18</sup>. Anashkin and coworkers<sup>19</sup> further supported this hypothesis by analysing the inhibition of *Desulfitobacterium hafniense* mPPase using three non-hydrolysable  $\text{PP}_i$  analogues (IDP, etidronate (ETD), and aminomethane bisphosphonate). Bisphosphonates, such as risedronate (RSD) and pamidronate (PAM), serve as primary drugs currently used to combat osteoclast-mediated bone loss<sup>20</sup>. Unlike IDP, which contains a P-N-P bond, bisphosphonates have a P-C-P bond, with its central carbon can accommodate up to two substituents, allowing a large compound variability. Therefore, understanding their inhibition mechanism on mPPases is crucial for developing future small molecule inhibitors.

Our previous work on serial time-resolved X-ray crystallography and electrometric studies on TmPPase directly observed structural asymmetry, where two monomers are in different states during  $\text{PP}_i$  hydrolysis upon the addition of substrate and  $\text{Na}^+$ , supporting a “pumping-before-



**Figure 1.**

### Inhibition effect of bisphosphonates on TmPPase.

**A.** Overall structure of the monomer TmPPase structure, consisting of hydrolytic centre, coupling funnel, ion gate and exit channel. **B.** Chemical structure of IDP and bisphosphonates and their inhibition activity against TmPPase. **C.** Activity of TmPPase modified with NEM (N-Ethylmaleimide) in the presence of  $MgCl$  (2.5 mM) and NaCl (20 mM) after incubation with  $Ca^{2+}$  (2 mM) and bisphosphonates (0.5 mM). Measurement was done in triplicate.

hydrolysis” energy coupling model<sup>7</sup>. However, except for the allosteric inhibitor ATC, which binds to a region near the exit channel, crystal structures of TmPPase bound to inhibitors at the active site are symmetric. To probe the proposed asymmetry caused by the inhibitor (and substrate) binding in solution, we employed double electron-electron resonance (DEER), also known as pulsed electron double resonance (PELDOR) spectroscopy. This method relies on the introduction of paramagnetic spin labels at selected protein residues, allowing for precise determination of electron-electron dipolar couplings and subsequently, inter-spin distances<sup>21-23</sup>, making it a powerful tool for probing the conformation and dynamics of integral membrane proteins<sup>24-28</sup>, including ion channels, transporters, outer membrane proteins, and receptors in their native environments<sup>29-35</sup>. As an ensemble technique, DEER can probe the presence of multiple conformational species including lowly-populated protein states, which are key to protein function<sup>36-38</sup>. Here, we solved two TmPPase structures in complex with ETD and zoledronate (ZLD) and monitored their conformational ensemble using DEER spectroscopy in solution. Overall, bisphosphonates can trigger conformational changes in the active site and near the exit channel of TmPPase in an asymmetric mode and under certain inhibitor-bound conditions; the DEER data are consistent with interspin distances predicted from an open/closed asymmetric model, and correlate with the corresponding X-ray structures. This, along with our electrometric studies detecting the Na<sup>+</sup> signal across the membrane, further suggests that ion pumping requires a fully closed state of one TmPPase monomer, supporting symmetry-breaking across the membrane, consistent with half-of-the-sites-reactivity<sup>7</sup>.

## Results

### Bisphosphonates are weaker TmPPase inhibitors than IDP

Bisphosphonates have been shown to inhibit mPPases<sup>19,39</sup>. To understand their binding mechanism to TmPPase, we first assessed the binding ability of seven distinct bisphosphonates to TmPPase by testing their inhibitory activity using the molybdenum blue assay (**Fig. 1B**), with IDP (IC<sub>50</sub> = 56 ± 5 μM) as a positive control<sup>40</sup>. Of the compounds tested, all the straight-chain primary amines (pamidronate (PAM), alendronate (ALE) and neridronate (NER)) had similar IC<sub>50</sub>s, ranging from 117 to 138 μM (*P* = 0.06). Substituting the -NH- of IDP with the -CCH<sub>3</sub>(OH)- of ETD resulted in a weaker IC<sub>50</sub> (> 200 μM). Similarly, branched aliphatic and aromatic bisphosphonates (ibandronate (IBD), ZLD and RSD) also showed weaker inhibition (IC<sub>50</sub> > 200 μM) (**Fig. 1B** and **Fig. EV1**).

To confirm that the binding of bisphosphonates to TmPPase induces conformational changes in the protein structure, we incubated the enzyme with the inhibitors and performed an *N*-ethyl maleimide (NEM) modification assay<sup>41</sup>. NEM covalently binds to exposed cysteine residues of the protein, forming a carbon-sulfur bond that can inhibit the protein activity if the residue is essential<sup>8</sup>. The binding of IDP has been reported to prevent the NEM modification of cysteine by reducing cysteine accessibility, thereby preserving TmPPase activity<sup>8</sup>. In the absence of inhibitors, NEM modification resulted in a decrease in TmPPase activity by approximately 40% (**Fig. 1C**), similar to the activity reduction observed with CaCl<sub>2</sub>, an inhibitor that binds to the open form of TmPPase<sup>8</sup>. Upon the addition of IDP, TmPPase adopts a closed conformation, rendering it resistant to NEM modification<sup>8</sup> (**Fig. EV2**); consequently, the enzyme remains largely unaffected by NEM. Although not as effective as IDP, all bisphosphonates prevent NEM modification to a comparable extent (**Fig. 1C**).

### TmPPase structures in complex with bisphosphonate inhibitors

To decipher the structural basis of bisphosphonates inhibition and their binding to TmPPase, we decided to solve their structures since all the bisphosphonates bound to TmPPase despite not being isosteres of PP<sub>i</sub> (**Fig. 1**). We obtained protein crystals for all the inhibitors, but they diffracted weakly, except for TmPPase in complex with ETD (TmPPase:ETD) and ZLD



(TmPPase:ZLD), which diffracted to resolutions of 3.2 Å and 3.3 Å, respectively. TmPPase:ETD crystallised in the presence of  $\text{Ca}^{2+}$ , which is a well-known mPPase inhibitor<sup>8</sup>, while TmPPase:ZLD crystallised without  $\text{Ca}^{2+}$ . Both data sets were anisotropic as analysed using the STARANISO server<sup>42</sup> (Table S1). We solved both structures by molecular replacement using the resting state structure (PDB ID: 4AV3) as the search model for TmPPase:ETD and the closed IDP-bound structure (PDB ID: 5LZQ) for TmPPase:ZLD. There were two molecules in the asymmetric unit for TmPPase:ETD and four for TmPPase:ZLD.

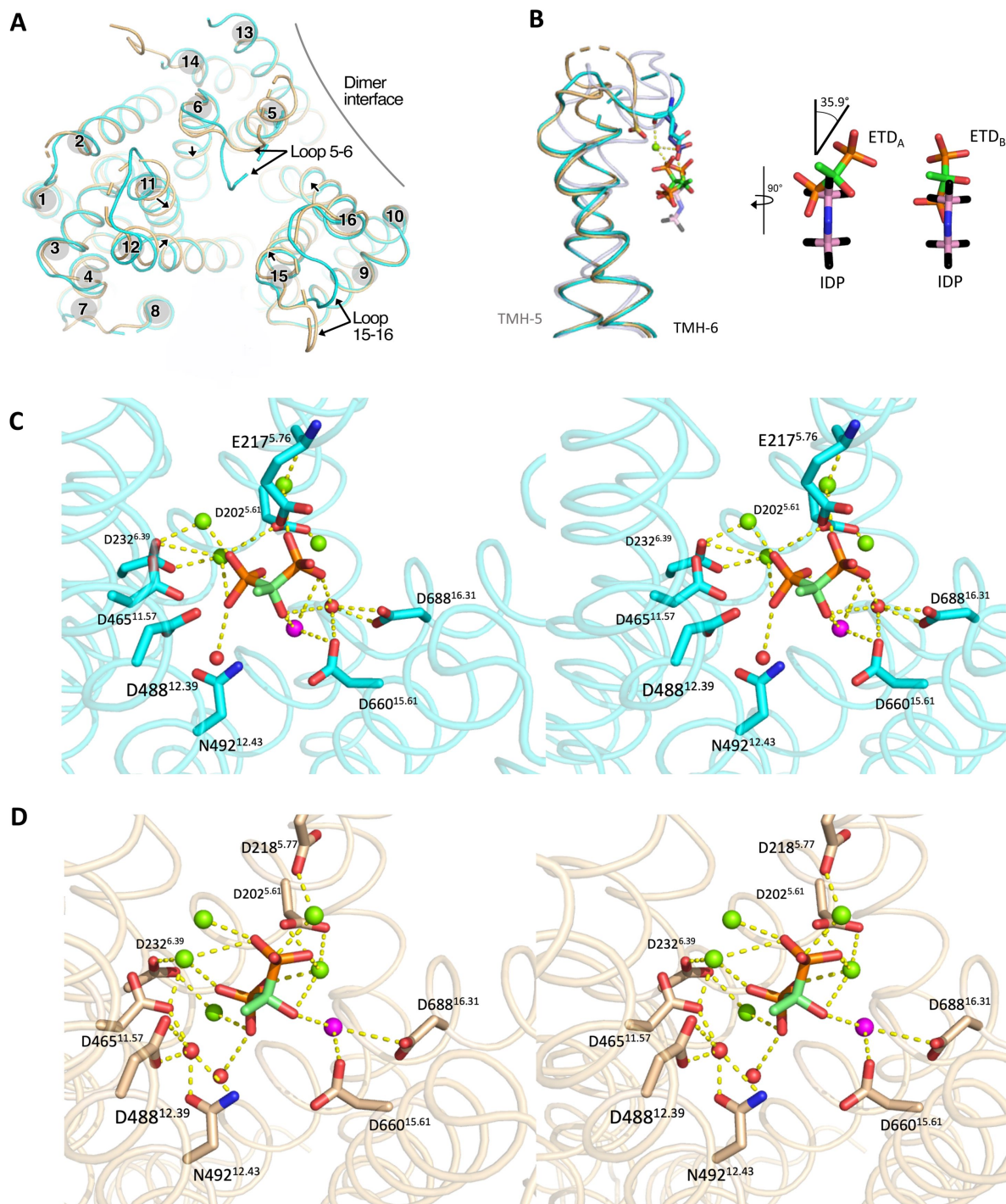
In the initial round of the refinement for the TmPPase:ETD structure, both chains displayed positive ( $F_o - F_c$ ) density at  $3\sigma$  in their hydrolytic centres that could accommodate ETD (Figs. EV3A-B, upper left panel). We also observed extra density that corresponds to a calcium ion in the resting state structure<sup>8</sup> (Figs. EV3A,B, upper left panel). Due to the high  $\text{Ca}^{2+}$  concentration (0.2 M) in the crystallisation condition, we placed the same ion at this position. After placing  $\text{Mg}^{2+}$  ions and water molecules in the difference density peaks, further rounds of refinement provided us with a reasonable  $2mF_o - F_c$  density map in the active site of both monomers (Fig. EV3A,B, right panel) and the POLDER (Omit) maps indicate a good fit of the compound to the density (Fig. EV3A,B, bottom left panel). Finally, the TmPPase:ETD structure was refined to an average resolution of 3.2 Å ( $h = 3.1$  Å,  $k = 3.6$  Å,  $l = 4.3$  Å) with the final  $R_{\text{work}}/R_{\text{free}}$  of 27.2% / 31.0 % (Table S1).

Similarly, the initial refinement of TmPPase:ZLD revealed positive ( $F_o - F_c$ ) density at  $3\sigma$  that could accommodate ZLD in all four chains in the asymmetric unit (Figs. EV4A-D, upper left panel). After placing  $\text{Mg}^{2+}$  ions and water molecules in the difference density peaks, further rounds of refinement provided us with a  $2mF_o - F_c$  density map in the active site for all monomers (Fig. EV4A-D, right panel) and was validated by POLDER (Omit) maps (Figs. EV4A-D, bottom left panel). The final refinement shows that the TmPPase:ZLD structure has an average resolution of 3.3 Å ( $h = 4.5$  Å,  $k = 4.2$  Å,  $l = 3.2$  Å) with a final  $R_{\text{work}}/R_{\text{free}}$  of 25.9 % / 30.4 % (Table S1).

## Asymmetry in the TmPPase complex with etidronate

Unlike the fully open TmPPase:Ca:Mg structure (PDB ID: 4AV3), there was additional density above the hydrolytic centre in both chains that could be fitted with several residues of loop5-6 (Fig. 2A and Fig. EV5). This left eight residues (V208<sup>5.67</sup>-L215<sup>5.74</sup>) in loop5-6 of chain A and three residues (L213<sup>5.72</sup>-L215<sup>5.74</sup>) in chain B unmodeled due to the lack of extra density. In the IDP-bound structure, these loops interact with IDP and form a tightly packed structured lid over the active site. However, in the TmPPase:ETD structure, despite interacting with ETD in both chains, these loops are positioned slightly above the active site (Fig. 2B-D), with loop5-6 of chain A extending more toward the centre compared to loop5-6 of chain B (Fig. 2A).

Structural alignment between chain A and B of TmPPase:ETD yields a root mean square deviation (RMSD) per Ca of 1.44 Å, approximately four times higher than the RMSD between chain A and B in the resting state (RMSD/Ca = 0.39 Å) (Table S2). Despite the overall structural similarity, further comparison between the monomers of TmPPase:ETD and those in the resting state revealed that chain B of TmPPase:ETD differs most from TmPPase:ETD chain A and from both chains in the resting state structure (Table S2). Notably, there are clear differences on the cytoplasmic (hydrolytic) side between the monomers of TmPPase:ETD; chain B adopts a slightly more constricted conformation than chain A (Fig. 2A). Bendix analysis<sup>43</sup> showed that three (TMH11, 12 and 15) out of six inner ring helices of chain B are more curved on the cytoplasmic side, bending towards the active site (Fig. EV6). Besides that, the loops on the cytoplasmic side of chain B (loops11-12, 13-14, and 15-16) appear to be more flexible, as indicated by more unresolved residues, compared to those in chain A (Fig. 2A). These observations suggest structural asymmetry between chains A and B in the TmPPase:ETD structure.



**Figure 2.**

### Structural asymmetry in the dimer active-site of TmPPase:ETD complex.

**A.** Top view of the superposition of chain A (cyan) and chain B (wheat) showing the relative movements (black arrow) of helices. **B.** Side view of the superposition of TMH5 and TMH6 in TmPPase:ETDs (chain A (cyan) and chain B (wheat)) and TmPPase:IDP (light blue; PDB: 5LZQ) showing the movement of loop5-6 and reorientation of ETD<sub>A</sub>, ETD<sub>B</sub> and IDP. The yellow dashed line shows the interaction of E217<sup>5.76</sup> of loop5-6 in Chain A with ETD and IDP, and D218<sup>5.77</sup> of loop5-6 in Chain B with ETD<sub>B</sub>; Close-up view of IDP superposed with ETD<sub>A</sub> and ETD<sub>B</sub>. **C.** Stereo representation (wall-eyed view) of residues in the active site with ETD coordinated (dashed lines), Ca<sup>2+</sup> (pink sphere) and nucleophilic water (red spheres) in a Mg<sup>2+</sup> metal cage (green spheres). **D.** Stereo representation (wall-eyed view) of residues in the active site with ETD<sub>B</sub> coordinated (dashed lines), Ca<sup>2+</sup> (pink sphere) and water (red spheres) in a Mg<sup>2+</sup> metal cage (green spheres).

The structural asymmetry arises because the binding of ETD<sub>A</sub> to monomer A induced conformational changes in monomer B, thereby affecting the binding pose of ETD<sub>B</sub> in monomer B. ETD comprises two phosphonate groups separated by a central carbon bonded to a hydroxyl group. Compared to the IDP location in the IDP-bound structure, ETDs are positioned above the IDP site, with the lower phosphonate group of ETDs located in the position of the upper (leaving-group) phosphonate group of IDP (**Fig. 2B**). However, the upper phosphonate group of ETDs in chains A (ETD<sub>A</sub>) and B (ETD<sub>B</sub>) is distinctly positioned; ETD<sub>A</sub> is tilted approximately by 35.9° relative to the IDP orientation, while ETD<sub>B</sub> is parallel to the IDP orientation (**Fig. 2B**). The lower phosphonate group position remains the same for both ETDs (**Fig. 2B**). As a result, loop5-6 of the two monomers is oriented differently. In chain A, this loop protrudes towards the active centre and E217<sup>5,76</sup> interacts with ETD via a Mg<sup>2+</sup> ion, while in chain B, the loop is more constricted and interacts with ETD via D218<sup>5,77</sup>, also mediated by a Mg<sup>2+</sup> ion (**Fig. 2B,C and D**). Furthermore, ETD<sub>A</sub> and ETD<sub>B</sub> interact with the active site via different residues (**Fig. 2C,D**). D465<sup>11,57</sup>, D488<sup>12,39</sup> and N492<sup>12,43</sup> in TMH11 and TMH12 were involved in the interaction with ETD<sub>B</sub> via a water molecule. Consequently, these two TMHs undergo slight inward movement, resulting in a more constricted conformation of chain B. Exchanging the ETD positions between the two protomers generated corresponding positive and negative difference electron density peaks, confirming distinct conformations of ETD within each protomer (**Fig. EV3C**). Nonetheless, the methyl group of ETDs in both chains points towards TMH12 (**Fig. 2C,D**), which might prevent complete closure of the hydrolytic centre and downward motion of TMH12.

## Structural distinction between zoledronate and IDP-bound TmPPase

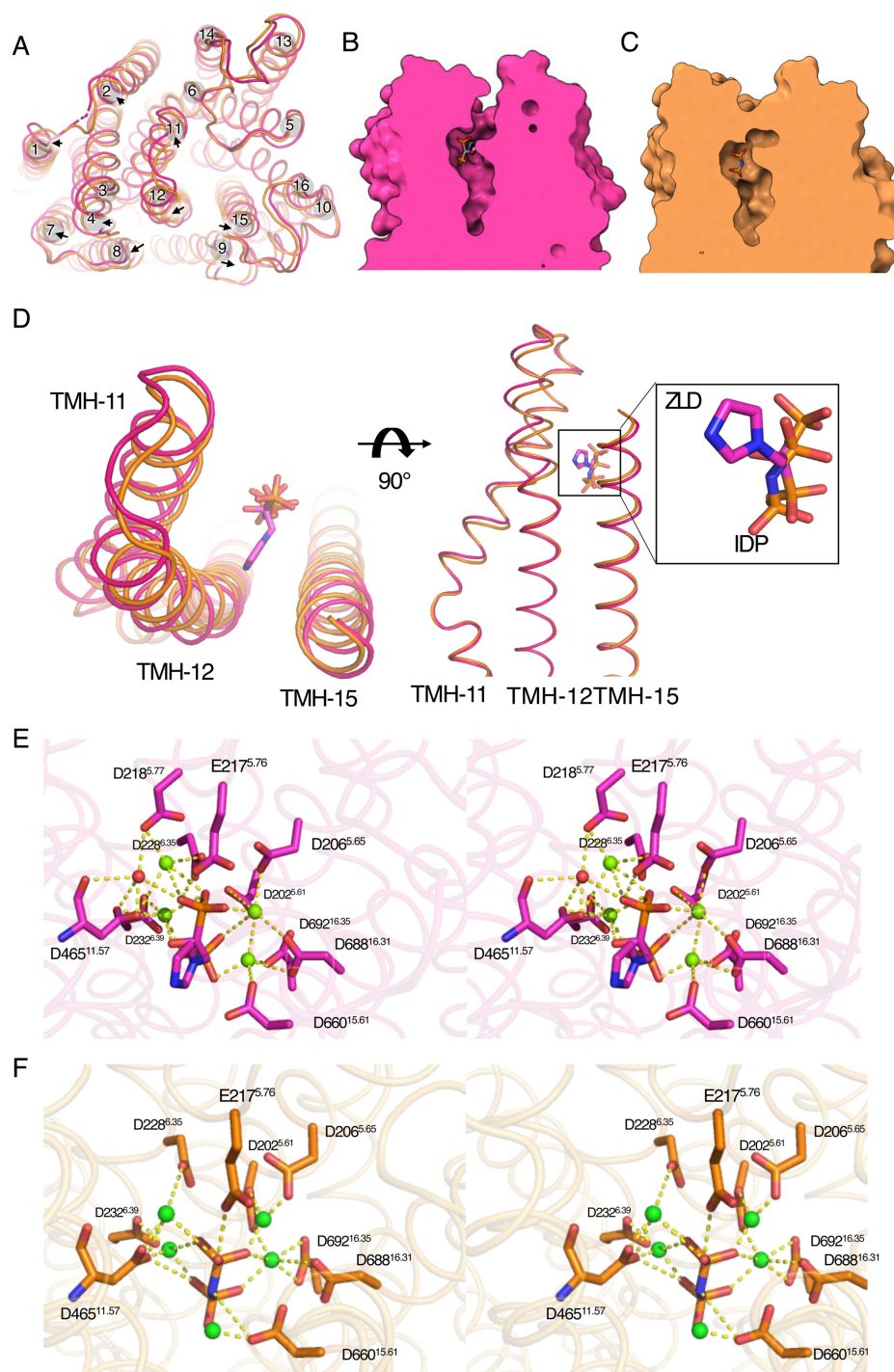
In contrast to the TmPPase:ETD structure, the TmPPase:ZLD structure adopts a partially closed conformation. The overall structure is more similar to the IDP-bound structure (RMSD/Cα of 0.760 Å) than the resting state structure (RMSD/Cα of 2.32 Å) (Table. S2). However, compared to the IDP-bound structure, the TmPPase:ZLD structure exhibits noticeable movements in three of six inner ring helices (TMH11, 12 and 15) and seven of ten outer ring helices (TMH1-4 and 7-9). These movements extend outwards from the hydrolytic centre (**Fig. 3A**), leaving it only partially closed. A cross-sectional view confirms this observation, showing the tunnel extending from the hydrolytic centre to the enzyme surface unlike in the IDP-bound structure, where it is closed (**Fig. 3B,C**). This is because ZLD is sterically bulkier than IDP due to the presence of the heteroaryl group, which points towards TMH11, 12, and 15 on the cytoplasmic side.

Although the hydrolytic centre of TmPPase:ZLD is more open, the coordination of the Mg<sub>4</sub>ZLD complex with the active site residues closely resembles that of Mg<sub>5</sub>IDP in the IDP-bound structure (**Fig. 3E,F**). ZLD is nonetheless positioned about 1.0 Å above IDP (**Fig. 3D**) because the steric bulk prevents it from sitting deeper into the hydrolytic centre. However, unlike the IDP-bound structure, and even though the arrangement of TMHs in the ion gate is almost identical, we did not observe any density for a Na<sup>+</sup> in the TmPPase:ZLD structure despite its higher resolution (*i.e.* 3.26 Å compared to 3.5 Å for the IDP-bound structure) (**Fig. EV7C**).

## Probing the solution-state conformational ensemble and dynamics of TmPPase by DEER spectroscopy

The X-ray structures of TmPPase with the different inhibitors bound to the active site show either a closed (TmPPase:IDP<sup>9</sup>), resting (TmPPase:Ca<sup>8</sup>) or asymmetric (TmPPase:ETD, **Fig. 2**) conformation. The asymmetric structure of the TmPPase with ETD is similar to that observed in our recent time-resolved study<sup>7</sup>. To probe the TmPPase conformational ensemble in solution under various inhibitor-bound conditions, we employed DEER spectroscopy. We selected three distinct sites (periplasmic side, S525; cytoplasmic side, C599; cytoplasmic side loop region, T211) on TmPPase, which were selectively spin-labelled with 2,5-dihydro-2,2,5,5-tetramethyl-3-[[[(methylsulfonyl)thio]methyl]-1H-pyrrol-1-yloxy (MTSSL, modification denoted as R1 hereafter) to





**Figure 3.**

### Comparison of the TmPPase:ZLD and TmPPase:IDP structures in the active site.

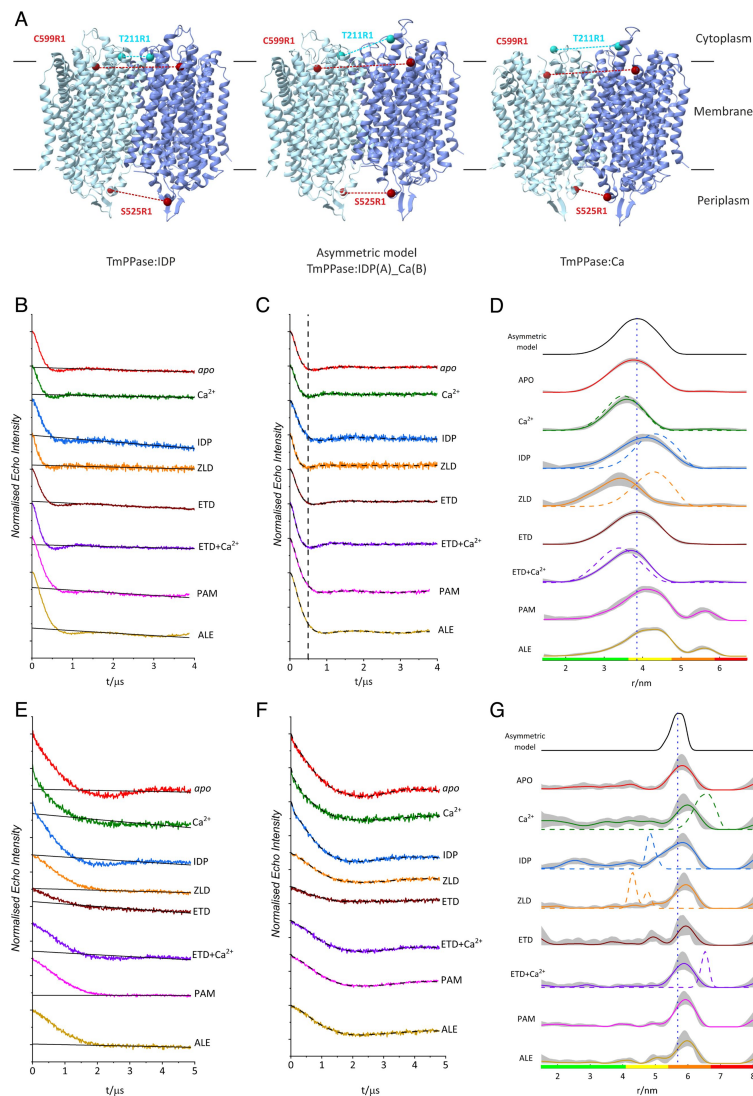
**A.** Top view of the superposition of TmPPase:ZLD (chain A, pink) and TmPPase:IDP (chain A, orange) (PDB: 5LZQ). Helices movements are indicated by a black arrow. **B** Cross-section side view of the active site in TmPPase:ZLD. **C** Cross-section side view of the active site in TmPPase:IDP. **D.** Top view of the superposition of TMH11, TMH12 and TMH15 in TmPPase:ZLD and TmPPase:IDP showed the movement of the hydrolytic centre and the orientation of ZLD and IDP. **E.** Stereo representation (wall-eyed view) showing the coordination of key residues in the active site with ZLD (dash line), and water (red sphere) in a  $Mg^{2+}$  metal cage (green spheres). **F.** Stereo representation (wall-eyed view) showing the coordination of key residues in the active site with IDP (dash line) in a  $Mg^{2+}$  metal cage (green spheres).

enable the measurement of interspin distances between the spin-labelled residue pairs. The selected sites were designed to capture the coupled gating transitions and conformational changes occurring on either side of the membrane (**Fig. 4A** [↗](#) and **Fig. EV8A** [↗](#)) without interfering with the activity of TmPPase. We achieved high mPPase spin labelling efficiency with no free (*i.e.* unbound or non-specifically bound) MTSSL being present, as evidenced by the continuous wave electron paramagnetic resonance (CW-EPR) spectra recorded at room temperature (**Fig. EV9** [↗](#)). CW-EPR spectra, which relates to the rotational correlation time, indicated that the spin label mobility increased sequentially from C599R1 to S525R1 and further to T211R1 across several tested conditions (*apo*, +Ca, +Ca/ETD, +ETD, +IDP *etc.*). This mobility trend aligns with the location of T211R1 on an exposed loop, which explains its higher mobility, whereas spin labelling of the more buried C599R1 required the addition of  $\text{Ca}^{2+}$  during sample preparation to induce partial structural opening. Unlike DEER, which provides insights into the long-range conformational changes of membrane proteins, CW-EPR offers information on the local environment of the spin label. The results show no significant difference in the local environment between the *apo* and inhibitor-bound state(s).

In addition, we generated *in silico* predictions of distance distributions for the three sites (S525R1, T211R1, and C599R1) using MtsslWizard<sup>44</sup> [↗](#) and ChiLife<sup>45</sup> [↗](#), based on the X-ray structures of TmPPase bound to different molecules (**Fig. 4** [↗](#) and **Fig. EV8** [↗](#)). In the case of T211R1, the X-ray electron density in loop<sub>A</sub>5-6 of the TmPPase:ETD (residues V208<sup>5,67</sup>-L215<sup>5,74</sup>; VGKTELNL) and TmPPase:Ca (residues T211<sup>5,70</sup>-R221<sup>6,28</sup>; TELNLPEDDPR) structures is missing, suggesting a highly dynamic or disordered state for this region. We therefore modelled this region using the Rosetta server<sup>46</sup> [↗](#) and used that to generate *in silico* distance distributions. These were overlaid with the experimentally derived DEER distance distributions (**Fig. 4D**, **G** [↗](#) and **Fig. EV8D** [↗](#)) for comparison. All T211R1 distance distributions were broad, consistent with the increased spin label mobility observed by CW-EPR, and the highly dynamic nature of these loop regions<sup>47</sup> [↗](#). Owing to the featureless raw DEER data recorded for 211R1 (**Fig EV8** [↗](#)), and broad distance distributions, we refrain from interpreting equilibria shifts based on this mutant. On the other hand, TmPPase dimers labelled at positions S525R1 and C599R1, located on opposite sides of the membrane, yielded high-quality DEER traces. Under all eight conditions tested (*apo*, +Ca, +Ca/ETD, +ETD, +IDP, +ZLD, +PAM, +ALE) for each site, strong dipolar oscillations were observed in the raw DEER data yielding robust distance distributions (**Fig. 4B**, **E** [↗](#)). This indicates that the modal distance shifts observed within the TmPPase ensemble are highly reliable. Both DeerAnalysis2022<sup>48</sup> [↗](#) and ComparativeDeerAnalyser 2.0<sup>49</sup> [↗](#) were used for background correction and regularisation of the dipolar traces, and their resulting distance distributions were in good agreement (**Fig. 4C**, **F** [↗](#) and **Fig. EV10** [↗](#)).

The separation of the S525R1 pair in the *apo* state (with no  $\text{Ca}^{2+}$  or inhibitor added) is broad with a modal distance of 3.8 nm (full width at half-maximum (FWHM) = 1.4 nm;  $\sigma$  = 0.60 nm) (**Fig. 4D** [↗](#)). In the presence of  $\text{Ca}^{2+}$ , the distance distribution is consistent with the predicted distances derived from the TmPPase:Ca structure, and the modal distance decreases (3.6 nm; FWHM = 1.0 nm;  $\sigma$  = 0.43 nm). In the presence of both  $\text{Ca}^{2+}$  and ETD (+Ca/ETD), we observe a similar modal distance (3.7 nm; FWHM = 1.2 nm;  $\sigma$  = 0.51 nm) to that of the *apo* and  $\text{Ca}^{2+}$  conditions, and the distribution is consistent with the predicted distance for the TmPPase:ETD structure (which corresponds to the +Ca/ETD condition). Furthermore, in the presence of ETD but no  $\text{Ca}^{2+}$ , the modal distance between the S525R1 pair on the different monomers increases to 3.9 nm (FWHM = 1.4 nm;  $\sigma$  = 0.60 nm). Although these shifts are relatively small, under favorable conditions DEER has the resolution to discriminate minute helical motions<sup>27,50</sup> [↗](#)–<sup>53</sup> [↗](#). The concerted nature of the modal distance shifts with respect to multiple different conditions at a single labelling site strongly suggests that preferential rotamer orientations are not the cause.

Upon visual inspection of the time-domain data (**Fig. 4C** [↗](#)), the first minimum of the dipolar oscillation, as indicated by the black dashed lines depicted for the *apo* state, shifts to shorter time (*i.e.*, higher frequency; shorter distance) for the TmPPase+ $\text{Ca}^{2+}$  condition, and to longer time (*i.e.*,



**Figure 4.**

### DEER distance distributions of TmPPase S525R1 and C599R1 under different conditions.

**A.** Symmetric structures (TmPPase:IDP (PDB: 5LZQ) and TmPPase:Ca (PDB: 4AV3) and asymmetric model (TmPPase:IDP(A)\_Ca(B)) of TmPPase. The sites mutated and labelled with MTSSL are shown as spheres, with T211R1 (Cyan) and C599R1 (maroon) on the cytoplasmic side (top) and S525R1 (maroon) on the periplasmic side (bottom) of the membrane. Distances between spin pairs are indicated as dashed lines, consistent with sphere colouring. DEER data of T211R1 is shown in supplementary Figure EV8. **B** and **E.** DEER raw data traces for S525R1 and C599R1, respectively. Each condition is labelled, and the raw data are colour-coded, with the background function indicated as solid black lines. **C** and **F.** DEER background-corrected time-domain traces for S525R1 and C599R1, respectively. The vertical black dashed line represents the minimum of the first oscillation in the apo state and aids visualisation to highlight the shifts in the oscillation minimum under different conditions. **D** and **G.** Distance distributions of S525R1 and C599R1, respectively. The in silico distance distribution corresponding to each spin pair modelled onto the asymmetric hybrid structure (TmPPase:IDP(A)\_Ca(B)) is shown at the top as a solid black line, with the modal distance shown as a vertical dashed line. In silico predicted distance distributions for each condition, modelled using the solved structures (TmPPase:Ca, TmPPase:Ca:ETD (PDB 9G8K), TmPPase:ZLD (PDB 9G8J), and TmPPase:IDP) are presented as coloured dashed lines overlaying the experimental distributions. The shaded regions represent the 95% ( $2\sigma$ ) confidence interval of the distributions, and the colour bars represent an assessment of the reliability of the distributions. The probability density within the green region indicates the mean distance, width, and peak shape are all reliable; the probability density within the yellow region indicates the mean distance and width are reliable; the probability density within the orange region indicates that the mean distance is reliable; the probability density within the red region indicates no quantitation is possible.



lower frequency; longer distance) for the TmPPase+ETD condition, recapitulating the trends observed in the distance domain. Interestingly, upon the addition of IDP, the resulting distribution has modal distance of 4.0 nm, (FWHM = 1.4 nm,  $\sigma$  = 0.60 nm); shorter than the predicted distance for the TmPPase:IDP structure (4.3 nm). Meanwhile, with the addition of PAM and ALE, the resulting distributions have modal distances (+PAM: modal distance = 4.1 nm, FWHM = 1.2 nm,  $\sigma$  = 0.51 nm; +ALE: modal distance = 4.3 nm, FWHM = 1.3 nm,  $\sigma$  = 0.55 nm) similar to the *in silico* distance distribution predicted from the TmPPase:IDP X-ray structure. In contrast, the addition of ZLD results in the shortest modal distance observed for the S525R1 pair, of 3.4 nm (FWHM = 1.2 nm,  $\sigma$  = 0.51 nm). Remarkably, this differs substantially from the *in-silico* distance distribution predicted from the X-ray structure of TmPPase:ZLD (4.3 nm), which is expected to be highly similar to that of TmPPase:IDP (RMSD/Ca = 0.571 Å) (see Discussion).

For the C599R1 dimer, the modal distance observed for all distributions under the tested conditions is approximately 5.8 nm (*apo*: modal distance = 5.8 nm, FWHM = 0.80 nm,  $\sigma$  = 0.34 nm; +Ca: modal distance = 6.0 nm, FWHM = 0.80 nm,  $\sigma$  = 0.34 nm; +IDP: modal distance = 5.8 nm, FWHM = 1.2 nm,  $\sigma$  = 0.51 nm; +ZLD: modal distance = 5.9 nm, FWHM = 0.80 nm,  $\sigma$  = 0.34 nm; +ETD: modal distance = 5.9 nm, FWHM = 0.70 nm,  $\sigma$  = 0.30 nm; +ETD/Ca: modal distance = 5.9 nm, FWHM = 0.80 nm,  $\sigma$  = 0.34 nm; +PAM: modal distance = 5.9 nm, FWHM = 0.70 nm,  $\sigma$  = 0.30 nm; +ALE: modal distance = 6.0 nm, FWHM = 0.70 nm,  $\sigma$  = 0.30 nm); (Fig. 4G); this is longer than the predicted 4.8 nm distance derived from the TmPPase:IDP structure – where both monomers are closed – but significantly shorter than the predicted 6.6 nm distance for the TmPPase:Ca and TmPPase:ETD structures, where both monomers are open. This deviation between prediction and experiment could be explained by the dimer adopting an asymmetric conformation under the physiological conditions used for DEER, with one monomer in a closed state and the other in an open state. To investigate the asymmetric arrangement between two TmPPase monomers, we combined chain A of the TmPPase:IDP structure with chain B of the TmPPase:Ca structure to generate an asymmetric model, termed TmPPase:IDP(A)\_Ca(B). We refer to the conformation of the TmPPase:IDP structure as ‘closed’ at both sides, even for residues not in the active site, for residues as in the TmPPase:Ca structure as ‘open’ at both sides. Our asymmetric model has, for instance, S525(A) ‘closed’ but S525(B) ‘open’.

The asymmetric model predicts a distance distribution that agrees closely with the DEER data obtained for the majority of the eight conditions tested for both C599R1 and S525R1 pairs (Fig. 4D and G). The distribution predicted by the asymmetric model also falls between the two conformational extremes (fully closed and fully open states) of TmPPase structures. To further delineate the best-fitting model of the S525R1 DEER data, particularly given their smaller range from 3.6-4.0 nm, which resembles both asymmetric (*i.e.* closed-open) and *apo*-state (*i.e.* open-open) models, Bhattacharyya coefficients<sup>54</sup> were calculated for the two models. The values are as follows: +Ca = 0.98 (*apo* model), 0.90 (asymmetric model); +IDP = 0.97 (*apo* model), 0.98 (asymmetric model); +ETD = 1.0 (*apo* model), 0.97 (asymmetric model); +ZLD = 0.95 (*apo* model), 0.84 (asymmetric model); +Ca/ETD = 0.98 (*apo* model), 0.91 (asymmetric model). It was not feasible to calculate these coefficients for the S525R1 +PAM and +ALE conditions, owing to being recorded on a different instrument, with a different x-axis, which was also the case for the C599R1 dataset. These coefficients for S525R1 indicate that the *apo*-state (*i.e.* open-open) model describes the experimentally derived distributions better for +Ca, +Ca/ETD, +ETD, and +ZLD, whereas the asymmetric (*i.e.* closed-open) model better describes the experimental data for +IDP. Higher Bhattacharyya coefficient values (closer to unity) signify better overlap (here taken as a proxy for model agreement). The ramifications of these calculations are further elaborated in the discussion.

## Effect of ETD and ZLD on sodium transport of TmPPase

Previously, we showed that IDP can facilitate a single Na<sup>+</sup> pumping cycle without hydrolysis<sup>7</sup>. To investigate whether pumping also occurs in the presence of ETD and ZLD, we recorded electrometric data during PP<sub>i</sub> hydrolysis and after binding of IDP, ETD and ZLD. In electrometric measurements, also known as solid-supported membrane-based electrophysiology<sup>55</sup>, a current

signal is generated and recorded when  $\text{Na}^+$  is transported across the membrane by the active reconstituted TmPPase. A maximal positive signal of  $0.6 \pm 0.03$  nA was detected within 0.15 ns (excluding instrument dead time) after the addition of 100  $\mu\text{M}$  substrate  $\text{K}_4\text{PP}_i$  (**Fig. 5A**). Most of the signal decayed within 1 second after  $\text{K}_4\text{PP}_i$  was added. Full signal recovery required several minutes before a repeat measurement could be performed on the same sensor. As expected, when 200  $\mu\text{M}$   $\text{K}_2\text{HPO}_4$  was added as a negative control, there was no signal, indicating that no ion pumping had occurred. Replacing the substrate with IDP resulted in a signal about half that of  $\text{K}_4\text{PP}_i$ . However, in the presence of 50  $\mu\text{M}$  ETD or 50  $\mu\text{M}$  ZLD, the signals were barely detectable, indicating no  $\text{Na}^+$  pumping was observed.

This observation is consistent with the DEER data described above and with the TmPPase:ETD and TmPPase:ZLD structures, where there is no density for  $\text{Na}^+$  in the ion gate. Interestingly, in all solved TmPPase structures,  $\text{Na}^+$  has been observed at the ion gate only in the IDP-bound structures (**Fig. 5B-E** and **Fig. EV7**). In the IDP-bound structure, four key residues (D703<sup>16.46</sup>, D243<sup>6.50</sup>, S247<sup>6.54</sup> and E246<sup>6.53</sup>) in the ion gate constitute the  $\text{Na}^+$  binding site (**Fig. 5E**). The formation of the site is driven by the downward motion of TMH16 (**Fig. EV7A**), transitioning from the resting state (TmPPase:Ca) to the closed state (TmPPase:IDP). The orientation of D703<sup>16.46</sup> of the TmPPase:ETD structure resembles the structure of TmPPase:Ca, rotated away from the  $\text{Na}^+$  binding site, causing a loss of  $\text{Na}^+$  binding (**Fig. 5B,C**). In the TmPPase:ZLD structure, D703<sup>16.46</sup> and K707<sup>16.50</sup> are oriented relatively similarly to the  $\text{Na}^+$  binding position in the TmPPase:IDP structure (**Figs. 5D,E** and **Fig. EV7C**), but no  $\text{Na}^+$  density was observed despite the higher resolution compared to the TmPPase:IDP structure (3.26 Å compared to 3.5 Å for the IDP-bound structure). This might be because the inhibitor restricts the complete closure of the active site and full constriction and downward movement of the inner helices (especially TMH12 and 16) (**Fig. 3A-D**), which hinder the  $\text{Na}^+$  pumping.

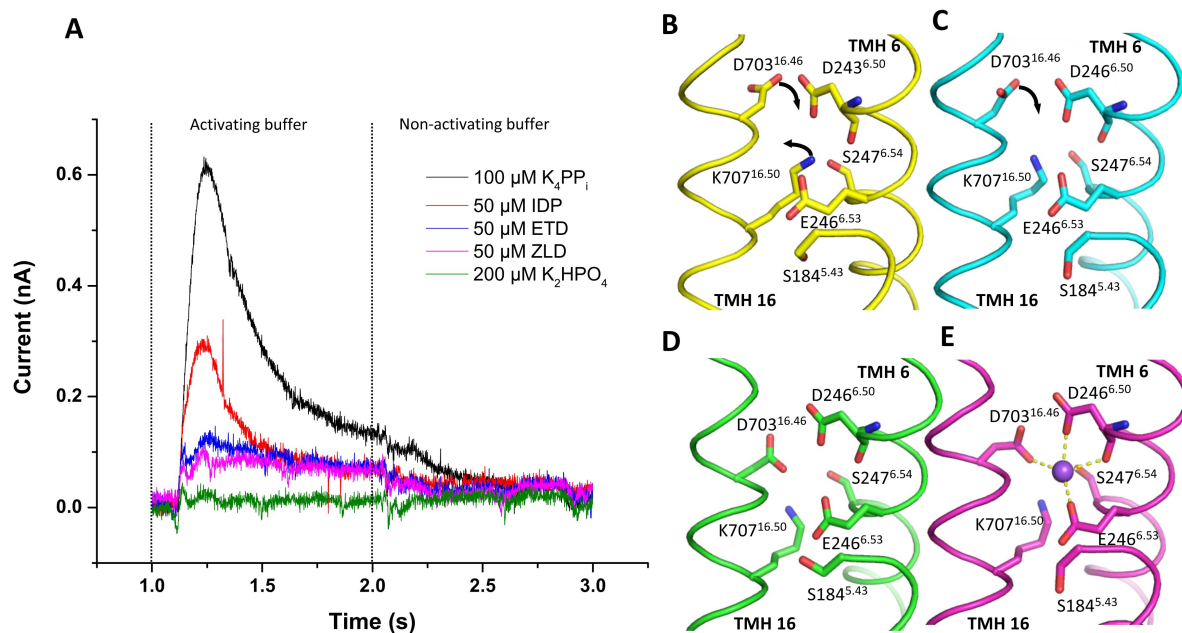
## Discussion

### Inhibition of TmPPase by bisphosphonates

The seven distinct bisphosphonates we tested exhibited varying levels of inhibition against TmPPase (**Fig. 1B**). ETD and IBD exhibited higher  $\text{IC}_{50}$  values compared to PAM, ALE, and NRD ( $p < 0.0001$ ) (**Fig. 1B**). This is consistent with the  $K_i$  of ETD on DhPPase (mPPase of *Desulfitobacterium hafniense*), which is approximately 67 times higher than that of amino methylene diphosphonate (AMDP), as measured by Viktor *et al.*<sup>19,56</sup>. The difference may be due to the introduction of an amino group in the side chain and its length<sup>39</sup>. Substituting the hydrogen (in ETD) with the benzene ring (in ZLD and RSD) decreases inhibitory activity ( $\text{IC}_{50} > 200$   $\mu\text{M}$ ). Nonetheless, these aromatic-containing compounds are still capable of preventing NEM modification on TmPPase (**Fig. 1C**), as further supported by the solved structure of ZLD bound with TmPPase (see Discussion section below).

### Catalytic asymmetry in mPPase

Some evidence for asymmetry in mPPase gating has been shown previously by kinetic studies<sup>7,19</sup> and captured in the time-resolved 600s and 3600s structures of TmPPase:PP<sup>7</sup>, where in both structures, one chain is in the open state (*i.e.* as in the *apo* structure) and the other is in the closed state (*i.e.* as in the IDP-bound structure). Our DEER data reveal clear differences in the binding of different inhibitors leading to a variety of open-closed states: IDP generates a closed-open state on both sides of the membrane, consistent with the presence of  $\text{Na}^+$  in the ionic gate and pumping, while ETD and ZLD generate a closed-open state on the cytoplasmic side, but an open-open states on the periplasmic side. These begin to explain the conformation changes upon substrate/inhibitor binding.



**Figure 5.**

### Transient currents of TmPPase $\text{Na}^+$ pumping and ion gate of TmPPase structures.

**A.** Curve of  $\text{Na}^+$  pumping current triggered by 100  $\mu\text{M}$  of  $\text{K}_4\text{PP}_i$ , 50  $\mu\text{M}$  of IDP, 50  $\mu\text{M}$  of ETD, 50  $\mu\text{M}$  of ZLD, and 200  $\mu\text{M}$  of  $\text{K}_2\text{HPO}_4$ . The vertical black dashed line represents the addition of activating buffer and non-activating buffer. **B-E.** Ion gate of TmPPase:Ca (yellow); TmPPase:ETD (cyan); TmPPase:ZLD (green); TmPPase:IDP (purple). The black arrows show the movement of residues of D703<sup>16.46</sup> and K707<sup>16.50</sup>.

In our study, the TmPPase:ETD structure captured the asymmetric binding of ETD (**Fig. 2** [↗](#)). Loop5-6, which interacts with ETD, moves inward to partially close the active site, but not as deeply as observed in the TmPPase:IDP structure (**Fig. 2A, B** [↗](#)). Here, ETD is positioned above the hydrolytic center (**Fig. 2B** [↗](#)) and cannot descend further due to the presence of  $\text{Ca}^{2+}$  in the active site (**Fig. 2C, D** [↗](#)), similar to the TmPPase:Ca structure (PDB ID: 4AV3). Thus, although ETD induces partial closure of loop5-6 and provides some stabilization, the overall arrangement of the inner and outer helices remains more like the open state rather than the fully closed state (**Fig. 2A** [↗](#)).

The DEER data on C599R1 provided reliable DEER distance distributions, and across all eight conditions tested, supported an asymmetric binding mode of compounds to TmPPase at the cytoplasmic side (**Fig 4E-G** [↗](#)). The modal distance of 5.8 nm differs significantly from the predicted C599R1 modal distance in the TmPPase:Ca (6.8 nm) and TmPPase:IDP (4.8 nm) structures. The presence of a minor population at approximately 5 nm observed in the presence of IDP or ETD is consistent with the predicted distance for the TmPPase:IDP structure, where both monomers are in a fully closed conformation. The 5.8 nm major peak corresponds to the closed-open conformation for IDP and ETD. Consequently, the DEER data on the cytoplasmic side demonstrate an equilibrium of at least two states: a minor population with IDP/ETD bound to both active sites, leading to a fully closed conformation on the cytoplasmic side, and a major population with IDP/ETD only bound to one active site; yielding an asymmetric closed-open state. This corresponds to the observed mechanism of substrate inhibition<sup>7,10</sup>, where binding to both active sites (*i.e.* closed-closed) decreases the activity of the enzyme in comparison with the half-occupied open-closed state. Under the condition tested, we did not observe ZLD, PAM, and ALE bound to both active sites, probably due to the bulkiness of the compounds.

We cannot completely rule out the possibility that the monomers adopt a metastable intermediate state: in such a case, we would expect the distance changes reported by DEER to be symmetric across both membrane sides. However, we observe symmetry breaking between the cytoplasmic and periplasmic TmPPase sites. Indeed, DEER data yield distance distributions similar to that of the hybrid asymmetric structure under all conditions (*apo*, +Ca, +Ca/ETD, +ETD, +ZLD, +IDP, +PAM, +ALE). The distance distribution for S525R1 (loop12-13) in the exit channel changes more between different conditions than C599R1 on the cytoplasmic side (**Fig 4** [↗](#)). Under +Ca, and +Ca/ETD conditions, its distance distribution remains largely unchanged, with a mean distance of ~3.5 to 3.7 nm (**Fig. 4D** [↗](#)), which is consistent with the predicted distance derived from their corresponding crystal structures. This suggests that conformational differences on the cytoplasmic side between the DEER data and crystal structures are not significantly manifested at the exit channel.

In the presence of IDP, however, we observed a longest distance distribution (~4.0 nm), consistent with the predicted distance from the hybrid asymmetric TmPPase:IDP(A)\_Ca(B) (**Fig 4A** [↗](#)) (closed-open), but neither the open-open nor closed-closed states. The ETD distance is intermediate, at ~3.9 nm (**Fig. 4D** [↗](#)), suggesting that a complete change to the closed conformational state on the periplasmic side does not occur, consistent with absence of  $\text{Na}^+$  in the exit channel. In contrast, with ZLD bound, the DEER distance distribution is the shortest (3.4 nm) (**Fig. 4D** [↗](#)), and significantly deviates from the predicted distance for TmPPase:ZLD structure. This discrepancy may arise because, in solution, while ZLD can enter the active site, its bulky heteroaryl group, which orients towards TMH 12 (**Fig. 3D** [↗](#)), prevents the full downward movement of this helix. This structural restriction results in a shorter DEER distance distribution. For PAM and ALE, the DEER distance distributions are even longer than those observed for IDP, closely matching the TmPPase:IDP structure. Since we do not have structures for their complexes with TmPPase, their orientation in the active site remains unknown.



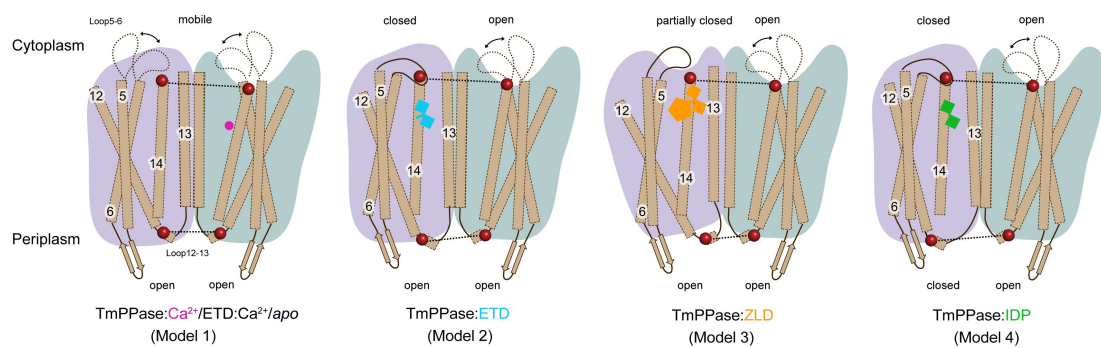
## Sodium ion pumping in TmPPase

Taken together, the X-ray crystallography and solution-state DEER data were used to propose a schematic for conformational transitions upon the addition of different compounds (**Fig. 6**). Model 1 represents an asymmetric state at the cytoplasmic side under *apo*, +Ca, and +Ca/ETD conditions. Loops5-6 are highly flexible, consistent with the broad distribution observed in DEER data for C211R1 and the missing electron densities in crystal structures. The periplasmic side remains in the ‘open’ state, with helices 12 and 16 ‘up’, consistent with the solved structures<sup>8</sup>. Model 2 describes the structural effects of ETD binding. C599R1, located at TMH14, reports a ‘closed-open’ state, with ligand binding to just one active site. However, there is no complete conformational change on the periplasmic side, the conformation is ‘open-open’. Model 3, with ZLD, features the bulky heteroaryl group of ETD pulling the TMH 12 away at the cytoplasmic side, further affecting its conformation at the periplasmic side in an ‘open-open’ state. Model 4, with IDP, which induces a ‘closed-open’ state at the cytoplasmic side, with ligand binding to just one active site, but also drives a full downward movement of TMH12 in one monomer. This conformational shift results in an asymmetric conformation at the exit channel, while the other monomer remains open, consistent with the ‘closed-open’ hybrid structure TmPPase:IDP(A)\_Ca(B) (**Fig. 4A**).

In a previous study<sup>7</sup>, we found that a single turnover event of Na<sup>+</sup> pumping only occurs in the presence of IDP. In our current Nanion SURFE<sup>2</sup>R experiment, we did not observe Na<sup>+</sup> pumping (**Fig. 5A**) upon the addition of ETD and ZLD, consistent with ETD and ZLD bound structures where no Na<sup>+</sup> was observed at the ion gate. These data are consistent with the models presented above (**Fig. 6**): IDP generates an asymmetric conformation in both the active site and in the exit channel, which occurs through the motion of TMH12. TMH5, TMH13 and TMH10 are key parts of intra-subunit communication between the two monomers<sup>7</sup>. (Loop12-13, where S525R1 is located, can be used to monitor the motion of TMH12.) However, neither ETD nor ZLD generate any Nanion SURFE<sup>2</sup>R signal; the structures with these ligands do not reveal Na<sup>+</sup> at the ionic gate. This is completely consistent with the C599R1 DEER distance distributions (see results), indicating that the cytoplasmic side (C599R1) is consistent with the ‘closed-open’ asymmetric conformation but that this has not propagated fully to the periplasmic side (S525R1), which is in the symmetric ‘open-open’ conformation, consistent with the Bhattacharyya coefficients, and which does not bind Na<sup>+</sup> at the ion gate. Consequently, the distance of 4.0 nm at S525R1, as observed in the IDP-bound sample, likely represents the minimal structural arrangement distance required for Na<sup>+</sup> pumping.

The DEER data thus provide a convincing structural explanation for why TmPPase is unable to pump Na<sup>+</sup> upon the addition of ETD or ZLD. In summary, EPR experiments in solution, coupled with new structures of inhibited forms of TmPPase, provide evidence supporting symmetry-breaking across the membrane, consistent with half-of-the-sites-reactivity<sup>7</sup>. In future studies, we will use time-resolved DEER to explore the order of conformational changes and how substrate addition is correlated with the release of product phosphate and ion pumping.

Note: During the revision of this manuscript, Baykov *et al*<sup>57</sup> published a stopped-flow analysis demonstrating that the proton pumping in mPPase from *Desulfotobacterium hafniense* only occurs in the presence of PP<sub>i</sub>, as measured by fluorescence changes in the pH-sensitive dye pyranine. In comparison, our Nanion SURFE<sup>2</sup>R can also detect signals induced by partial ion pumping or charged amino acid rearrangement, rather than solely ion pumping. The half reduction in signal in the presence of IDP may be due to Na<sup>+</sup> being translocated to the ion gate and locked there without further release, consistent with the TmPPase:IDP structure and our DEER data. The weak signals observed in the presence of ETD or ZLD are likely due to charged amino acid rearrangements induced by their binding.



**Figure 6.**

### Models based on DEER distance distributions for TmPPase S525R1 and C599R1.

Four DEER models showing major conformational ensembles of TmPPase in solution. Two monomers are colored purple and green, respectively. All TMHs are shown in brown; mobile loop5-6 is indicated by a black dashed line, while fixed loop5-6 and loop12-13 are indicated by a solid black line; The labelling sites are represented by maroon spheres. Ca<sup>2+</sup> is shown as a magenta circle; IDP is shown as purple squares; ETD as cyan squares connected by a cyan stick; ZLD as an orange pentagon.

## Materials and methods

### Protein expression and purification

TmPPase expression and purification have been described previously<sup>58,59</sup>. Briefly, pRS1024 plasmid containing his-tagged TmPPase was freshly transformed into *Saccharomyces cerevisiae* strain BJ1991. The cells were cultured in 250 ml of selective synthetic complete drop-out (SCD) media overnight before being added to 740 ml of 1.5' YP media with 2% glucose. The cells were then cultured for 8 h at 30 °C, collected by centrifugation (4,000 rpm, 10 min) and lysed at 4 °C using a bead beater with 0.2 mm glass beads. The membrane fraction was collected by ultracentrifugation (100,000 × g, 45 min) and the pellets were resuspended in buffer containing 50 mM MES-NaOH pH 6.5, 20% (v/v) glycerol, 50 mM KCl, 5.2 mM MgCl<sub>2</sub>, 1.33 mM dithiothreitol (DTT), 2 µg ml<sup>-1</sup> (w/v) pepstatin-A (Sigma) and 0.334 mM PMSF (Sigma). The membranes were solubilised in solubilisation buffer (50 mM MES-NaOH pH 6.5, 20 % (v/v) glycerol, 5.33 % (w/v) n-dodecyl-b-D-maltopyranoside (DDM) (Anatrace)) using the 'hot-solve' method<sup>59</sup> at 75 °C for 1.5 h. After centrifugation to remove denatured proteins, KCl (to a final concentration of 0.3 M) and 2 ml of Ni-NTA beads (Qiagen) were added and incubated at 40 °C for 1.5 h, and then loaded into an Econo-Pac® column (Bio-Rad). Then the column was washed with two column volume (CV) of washing buffer (50 mM MES-NaOH pH 6.5, 20% (v/v) glycerol, 50 mM KCl, 20 mM imidazole pH 6.5, 5 mM MgCl<sub>2</sub>, 1 mM DTT, 2 mg/ml (w/v) pepstatin-A, 0.2 mM PMSF and 0.05% DDM (Anatrace) and eluted with 2 CV of elution buffer (50 mM MES-NaOH pH 6.5, 3.5% (v/v) glycerol, 50 mM KCl, 400 mM imidazole pH 6.5, 5 mM MgCl<sub>2</sub>, 1 mM DTT, 2 mg/ml (w/v) pepstatin-A, 0.2 mM PMSF and 0.5% octyl glucose neopentyl glycol (OGNPG, Anatrace).

### TmPPase activity assay

TmPPase activity and bisphosphonates inhibition assay were performed using the molybdenum blue reaction method in 96-well plate format as reported previously<sup>40</sup>. Before the assay, the enzyme was reactivated by adding to the mixture of 30 mg/ml of soy-bean lecithin (Sigma) in 20 mM Tris-HCl pH 8.0 with 4.5% DDM and incubated at 55 °C for 15 min. The activity reaction was done in the reaction buffer (60 mM Tris-HCl pH 8.0, 5 mM MgCl<sub>2</sub>, 100 mM KCl, and 10 mM NaCl) and started by adding 2 mM Na<sub>4</sub>PP<sub>i</sub> at 71 °C for 5 min.

### NEM modification assay

The NEM modification assay was performed as reported previously with slight modification<sup>8</sup>. Briefly, 0.4 mg/ml of the reactivated TmPPase was mixed with the modification buffer (20 mM MES-KOH pH 6.5, 0.05% DDM, 2.4 mM MgCl<sub>2</sub>, 100 mM KCl, and 20 mM NaCl) and different inhibitors (2 mM CaCl<sub>2</sub>, 0.5 mM IDP, and 0.5 mM of bisphosphonates) and incubated on ice for 30 min. Afterwards, 100 mM *N*-ethyl maleimide (NEM) (Thermo Scientific) was added and the mixture was further incubated for 10 min. The NEM-modification reactions were stopped by adding 2 mM DTT and the residual activity of the enzyme was performed using the molybdenum blue reaction assay after removing excess inhibitors<sup>60</sup>.

### Crystallisation and structure determination

For co-crystallisation with bisphosphonates, the purified TmPPase was buffer exchanged to the crystallisation buffer (50 mM MES-NaOH pH 6.5, 3.5% (v/v) glycerol, 50 mM KCl, 5 mM MgCl<sub>2</sub>, 2 mM DTT and 0.5% OGNPG) on a Micro Bio-Spin 6 column (Bio-Rad) and then diluted to a concentration of 10 mg/ml. Prior to crystallisation, 1 mM bisphosphonates was added to the protein solution, incubated at room temperature for 30 min, and centrifuged for 20 min (16,000 g, 4 °C). Crystallisation trials were done using a Mosquito robot (SPT Labtech) by sitting drop vapour-diffusion method using MemGold™ screen (Molecular Dimensions) in MRC 2-well crystallisation

plates (Swissci), and the drops were monitored at 22 °C using the minstrel DT UV imaging system (Formulatrix). Crystal hits appeared on the MemGold™ screen under different conditions. Harvestable crystals appeared within several days and were frozen directly from the mother liquor. For the TmPPase cocrystallised with etidronate, the best diffracting crystal was observed from a solution containing 0.2 M CaCl<sub>2</sub>, 0.1 M HEPES pH 7.0, and 33% PEG400, while for TmPPase cocrystallised with zoledronate, the best diffracting crystal was observed from a solution containing 0.1 M MES pH 6.5, 0.1 M NaCl, 33% PEG400, and 4% ethylene glycol.

X-ray diffraction data were collected at Diamond Light Source (DLS) (UK) on the I03 (TmPPase:ETD) and I04-1 beamline (TmPPase:ZLD) at 100 K on a Pilatus 6M detector. The data were merged and scaled using X-ray Detector Software (XDS)<sup>61</sup> and the structure was solved by molecular replacement with Phaser<sup>62</sup> using the resting state (4AV3)<sup>8</sup> and IDP-bound (5LZQ) state<sup>9</sup> of TmPPase structure as the search model for TmPPase:Etidronate and TmPPase:Zoledronate, respectively. The structures were built and refined using phenix.refine<sup>63</sup> and Coot<sup>64</sup>. X-ray data and refinement statistics are listed in Table 1.

## EPR Spectroscopy

### Sample preparation for EPR spectroscopy

For EPR spectroscopy measurements, we utilized a nearly Cys-less construct, retaining only endogenous cysteine C183 due to its buried location and functional importance. Residue S525, located in the periplasmic loop12-13 of the TmPPase exit channel, was mutated to cysteine and covalently modified with a methanethiosulfonate thiol-specific spin label (MTSSL) to introduce a paramagnetic centre<sup>65,66</sup> (the labelled protein is referred to as S525R1). At the cytoplasmic side of the membrane interface, we constructed the TmPPase T211C variant, which is located in loop5-6 and above the active site (the MTSSL labelled mutant is referred to as T211R1). We also spin labelled an endogenous cysteine residue, C599, (after mutating back the S599 to cysteine) on the cytoplasmic transmembrane helix 14 (the labelled protein is referred to as C599R1).

The S525C, 599C, and T211C proteins were expressed as outlined above. The frozen cell pellets were lysed using cryo-milling (Retsch model MM400). 1 mM TCEP was used to replace DTT in the purification steps preceding spin labelling and the remaining purification was carried out as above. Each protein was spin-labelled with MTSSL while immobilised to the Ni-NTA resin (or mixed following Cys mutant elution) as previously described<sup>37,67</sup>. Briefly, for MTSSL labelling, MTSSL was added in spin-label buffer (20 mM MOPS-NaOH, 5 mM MgCl<sub>2</sub>, 50 mM KCl, 3.5% glycerol, 0.03% DDM at pH 7.5) at 10-fold molar protein excess and incubated for 2 hours at room temperature. For C599, 10 mM CaCl<sub>2</sub> was added in the buffer to increase the accessibility of the site for spin-labelling (i.e., to induce partial opening). Spin-labelled protein was eluted from the Ni-NTA resin column, concentrated and subsequently purified using size-exclusion chromatography using Superose 6 increase 10/300 GL (GE Healthcare) and equilibrated in 20 mM MES-NaOH, pH 6.5, 5 mM MgCl<sub>2</sub>, 50 mM KCl, 3.5% glycerol, 0.05% DDM. The eluted purified protein fractions were concentrated, buffer exchanged with buffer prepared in D<sub>2</sub>O and split into aliquots for incubation with a final concentration of 2 mM of all inhibitors or 10 mM CaCl<sub>2</sub> (30 min, RT). The protein activity was tested as described above, and the protein samples were tested for spin labelling by CW EPR spectroscopy, and then 40 % ethylene glycol-*d*<sub>6</sub> was added to each sample before flash freezing for DEER measurement.

### Continuous Wave EPR (CW-EPR) spectroscopy

CW EPR experiments were performed on a Bruker Magnetech ESR5000 X-band spectrometer (9.4 GHz). The spin-labelled sample was loaded into a 3 mm (o.d.) quartz EPR tube before the addition of ethylene glycol-*d*<sub>6</sub>. The samples were measured at room temperature (298 K), as TmPPase is

more thermally stable than most membrane proteins. The measurements were performed in a magnetic field range, 330-345 mT, with a 60 s sweep time, 0.1 mT modulation, 100 kHz frequency, and 10 mW (10 dB) microwave power.

## Double Electron-Electron Resonance (DEER, or PELDOR) spectroscopy

### DEER distance measurements and set-up

EPR recordings were collected as previously described<sup>68</sup> using a Bruker ELEXSYS E580 spectrometer operating at Q-band (34 GHz) frequency, equipped with a QT-II resonator in a cryogen-free variable temperature cryostat (Cryogenic Ltd.) with a temperature range 2-300 K. In brief, spin-labelled protein samples were prepared in 3 mm outer diameter quartz tubes and data was recorded at 50 K. The detection pulse sequence used was a refocused Hahn echo:  $\pi/2 - \tau_1 - \pi - \tau_1 - \tau_2 - \pi - \tau_2$  - echo, with  $\pi/2$  and  $\pi$  observer pulse lengths of 16 and 32 ns, and  $\pi$  inversion pulse lengths of 16-20 ns,  $\tau_1$  of 380 ns and  $\tau_2$  of 2000-5000 ns, depending on construct. Unless otherwise stated: the magnetic field and microwave frequency were adjusted for the maximum of the nitroxide spectrum to coincide with the pump pulse position, while the observer pulse was placed at either 65 MHz (for T211R1 and S525R1 measurements) or 80 MHz (C599R1 measurements) frequency offset. Measurements were recorded using either a 150 W (for T211R1 and S525R1 measurements) or a 300 W (for C599R1 measurements) travelling wave tube (TWT; Applied Systems Engineering). All pulses were generated using an integrated Bruker SpinJet AWG, and measurements were recorded using a 16-step phase cycle on the detection pulses was used to remove unwanted echo crossings<sup>69</sup>. Finally, electron-spin echo envelope modulation (ESEEM) arising from electron-nuclear coupling to deuterium was suppressed using an 8-step tau-averaging cycle<sup>70</sup>, with a time-increment of 16 ns.

For the measurements of S525R1 TmPPase (excluding the +PAM and +ALE conditions), a 4 ns dipolar increment to yield the DEER trace. Owing to significant excitation bandwidth overlap between observer and pump pulses at low frequency offset (-65 MHz), the presence of a “2+1” artefact exacerbated data treatment. Therefore, traces were recorded using a  $\tau_2$  of 5000 ns (and truncated to 4000 ns for data processing (see DEER data analysis and processing section below)), 16 shots-per-point, 647 points, and a shot repetition time (SRT) of 3060  $\mu$ s. Scans were recorded until a sufficient signal-to-noise ratio was obtained, typically with datasets averaged overnight. For the measurements of S525R1 TmPPase +PAM and +ALE, a 300 W travelling wave tube (TWT; Applied Systems Engineering) was used, operating at Q-band frequency. Traces were recorded with a 12 ns dipolar increment using a  $\tau_2$  of 4000 ns, 10 shots-per-point, 348 points, and SRT of 2000  $\mu$ s.

For the measurements of C599R1 TmPPase, a 12 ns dipolar increment was used, and traces were recorded using a  $\tau_2$  of 5000 ns, 10 shots-per-point, 432 points, and a SRT of 2000  $\mu$ s. Scans were recorded until a sufficient signal-to-noise ratio was obtained, typically with datasets averaged overnight. For all measurements of T211R1 (excluding the apo and +Ca/ETD measurements), a 4 ns dipolar increment was used, and traces were recorded using a  $\tau_2$  of 2000 ns, 32 shots-per-point, 432 points, and a SRT of 3000  $\mu$ s. Finally, for the apo and +Ca/ETD measurements, a 4 ns dipolar increment was used, and traces were recorded using a  $\tau_2$  of 4000 ns, 16 shots-per-point, 525 points, and a SRT of 3000  $\mu$ s.

### DEER data analysis and processing

Distance distributions were determined from the time traces using various methodologies as best practices to get reliable results and to ensure self-consistency<sup>71</sup>. In the present work, we used two different programs, DeerAnalysis2022<sup>71,72</sup> and ComparativeDeerAnalyzer2.0<sup>71</sup>, with results in the main text corresponding to the DeerAnalysis2022 processing. The S525R1 and T211R1



data recorded with a low frequency offset (65 MHz) yielded strong “2+1” artefacts, owing to overlapping pulse excitation profiles. To address this, all S525R1 data sets, and the apo and +Ca/ETD measurements for T211R1, were truncated or recorded to 4000 ns<sup>73</sup>, respectively, and then phase and background corrected using the ‘!’ automated adjustment. The background corrected traces were then transformed from the time-domain to the distance-domain using Tikhonov Regularization<sup>74</sup>, and the quality of the fit was assessed based on the L-curve criterion and the shape of the Pake pattern. The resultant background correction was then validated using a module for Tikhonov validation implemented in DeerAnalysis2022. The validation was carried out after initial Tikhonov regularization, varying the background start time from 5% to 80% of the respective time windows of the cut data for 16 trials. From this, the raw data were re-loaded and processed (Tikhonov regularization) with the cutoff and background start time as established from the first round of validation. This is the starting point for a full validation, where the background start time was again varied from 5% to 80% of the time window for 16 trials, as well as some added “white noise” with a level of 1.50 for 50 trials. The resulting validation trials were pruned and yielded the distance distribution and confidence interval. The ComparativeDeerAnalyser2.0 (CDA) was used to automate data processing and reduce operator bias. The corresponding output data for S525R1 and C599R1 TmPPase are shown in EV10.

Methods for B. coefficients calculation<sup>54</sup>: Bhattacharyya coefficients were used as similarity metric between experimental apo-state distribution and the in silico distribution predicted from the asymmetric hybrid structure. Following equation (1):

$$BC = \sum_{n \in N} \sqrt{P(n) \cdot Q(n)} \quad (1)$$

where  $P(n)$  and  $Q(n)$  are the normalised probability distributions (i.e. to convert from probability density distributions to probability distributions (2)) on the same domain  $N$ .

$$P(n) = \frac{P(n)}{\sum_{n \in N} P(n)} \quad (2)$$

## In silico spin labelling and modelling

MtsslWizard<sup>44</sup> and ChiLife<sup>45</sup> were used to predict in silico distance distributions for the T211R1, and S525R1 and C599R1 labelling sites of TmPPase, respectively. The coordinates of the respective X-ray structures were TmPPase:Ca PDB 4AV3, TmPPase:Ca:ETD PDB 9G8K, TmPPase:ZLD PDB 9G8J, and TmPPase:IDP PDB 5LZQ) for the different conditions were uploaded to the online MTSSL Suite server, labelled at T211 sites (both monomer A and B) with R1 and the rotamer cloud was generated using the “tight” labelling mode (i.e. zero steric clashes allowed). The PDB structures (including the asymmetric hybrid TmPPase\_Ca:TmPPase\_IDP structure) were also loaded into ChiLife and R1 sidechains were introduced, individually, at S525 and C599 sites. For consistency with MTSSLWizard predictions, the accessible volume approach to calculate rotamer clouds was used.

## Electrometric measurement

For the Nanion SURFE<sup>2</sup>R experiment, purified TmPPase was reconstituted into liposomes as previously described with some modifications<sup>9</sup>. Briefly, the purified protein was buffer exchanged into a reconstitution buffer (50 mM MOPS-KOH pH 7.2, 50mM KCl, 5mM MgCl<sub>2</sub>, and 2mM DTT) to remove Na<sup>+</sup> and glycerol and then diluted to 50 mg/ml concentration. 15 ml of liposome solution (120 mg/ml soy-bean lecithin in 50 mM MOPS-KOH pH 7.2) was mixed with 1 ml of diluted protein sample. SM-2 Bio-beads were added in increments to a final concentration of 0.25 mg/ml and then placed into a mixer at 4 °C for 6 h to ensure beads stayed in suspension. The

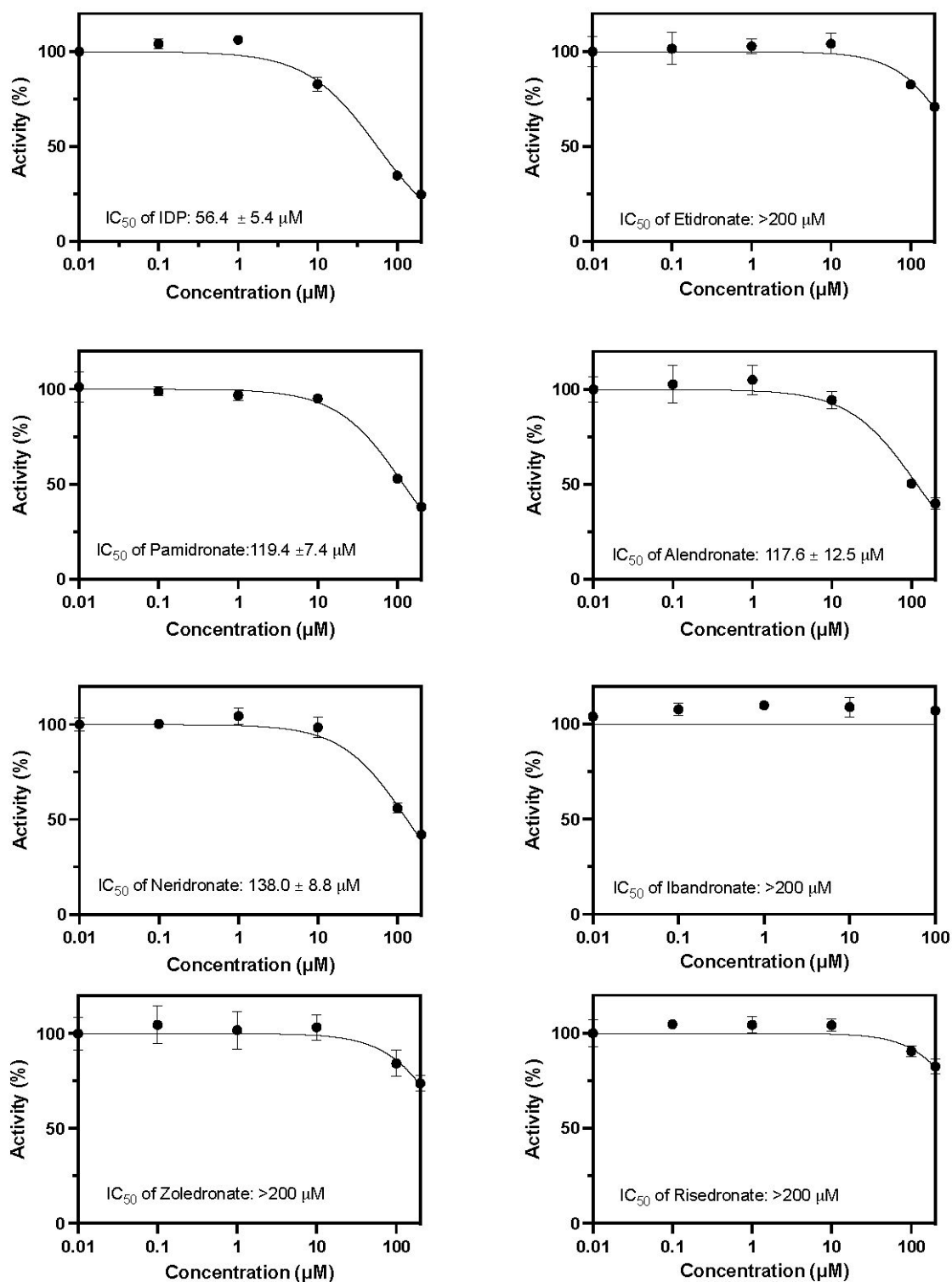
proteoliposomes were collected and frozen at -80 °C in aliquots. To ensure that the reconstituted protein was still active, the hydrolytic activity was performed using the molybdenum blue reaction assay<sup>60</sup> [↗](#).

Electrometric measurements were performed on a SURFE<sup>2</sup>R N1 instrument from Nanion Technology (Munich, Germany). The gold sensors were prepared based on the 'SURFE<sup>2</sup>R N1 protocol', including thiolation of the sensor surface and assembly of the lipid layer using sensor prep A2 and B solutions. 15 ml of sonicated proteoliposomes, followed by 50 ml of the rinsing buffer (50 mM MOPS-KOH pH 7.2, 50 mM NaCl, 5 mM MgCl<sub>2</sub>) were applied directly to the sensor surface. Sensors were centrifuged for 30 minutes at 2500 g and incubated at 4 °C for 3 h. The sensors were mounted in the SURFE<sup>2</sup>R N1 and rinsed once with 1 ml of rinsing buffer (50 mM MOPS-KOH, pH 7.2, 50 mM NaCl, 5 mM MgCl<sub>2</sub>). Measurements were performed for 3 s by consecutively flowing non-activating buffer B (50 mM MOPS-KOH pH 7.2, 50 mM NaCl, 5 mM MgCl<sub>2</sub>, 200 μM K<sub>2</sub>HPO<sub>4</sub>) and activating buffer A (50 mM MOPS-KOH, 50 mM NaCl, 5 mM MgCl<sub>2</sub>) containing substrate (100 mM K<sub>4</sub>PP<sub>i</sub>) or inhibitors (50 mM IDP, 50 mM ETD or 50 mM ZLD) across the sensor for 1 s each in a BAB sequence. Charge transport across the membrane is initiated by substrate or inhibitor in buffer A, which flows across the sensor between 1 and 2 s. The transport of positively charged ions during this period results in a positive electrical current, the signal output of the SURFE<sup>2</sup>R N1 instrument. Between each measurement, the sensor was washed with 1 ml rinsing buffer and incubated for 60 seconds. The measurements were tested in triplicates.

## Data availability

The atomic coordinates and structure factors of the TmPPase:Etidronate and TmPPase:Zoledronate complex have been deposited in the Protein Data Bank, [www.rcsb.org](http://www.rcsb.org) [↗](#) (PDB ID: 9G8K and 9G8J).

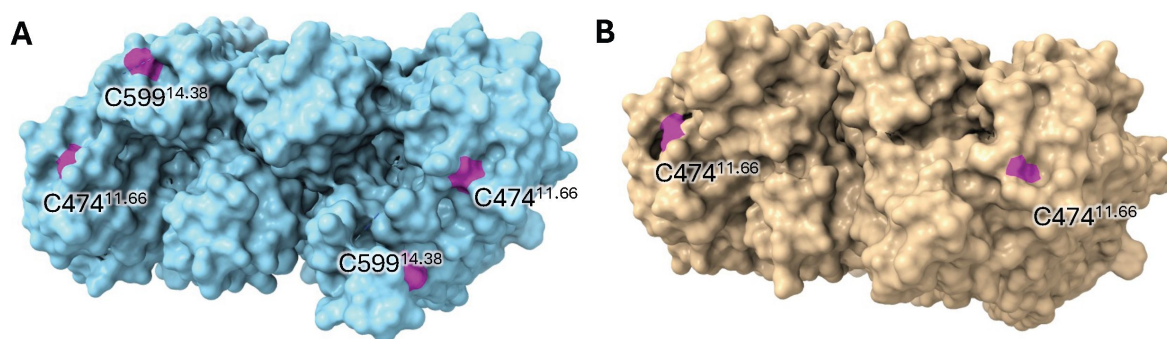
## Supplementary figures and tables



**Fig EV1.**

#### **Inhibition of TmPPase by bisphosphonates.**

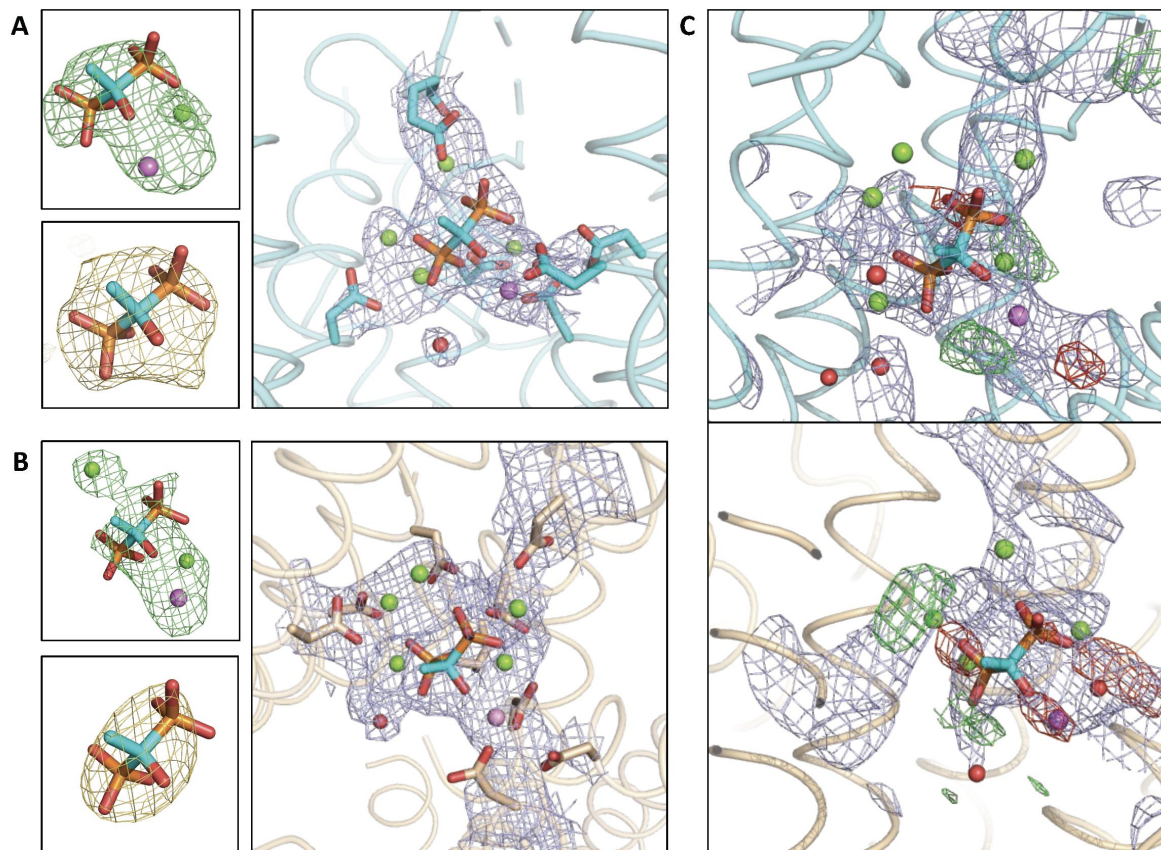
All data are shown as mean  $\pm$  SD with three replicates.



**Fig EV2.**

**Top view of accessible cysteines for NEM modification.**

**A.** Two exposed cysteines in both monomers of TmPPase:Ca (PDB: 4AV3; cyan). **B.** One exposed cysteine in both monomers of TmPPase:IDP (PDB: 5LZQ; wheat).

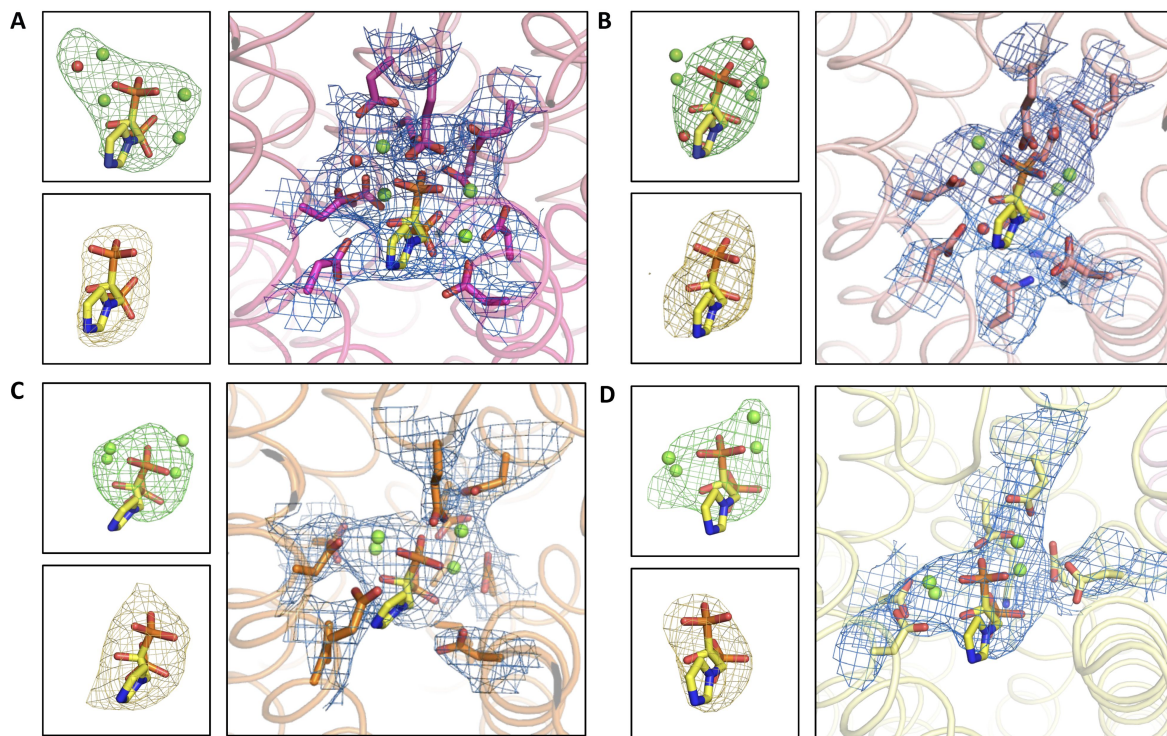


**Fig EV3.**

### Electron density maps of ETD at the active sites.

**A.**  $mF_o-F_c$  omit map with positive density of the  $ETD_A$  shown in green mesh (*Top left*); Polder omit map of  $ETD_A$  (*Bottom left*) shown in yellow mesh;  $2mF_o-F_c$  (light blue mesh) map of  $ETD_A$ , ions and surrounding residues (*Right*). **B.**  $mF_o-F_c$  omit map with positive density of the  $ETD_B$  shown in green mesh (*Top left*); Polder omit map of  $ETD_B$  (*Bottom left*) shown in yellow mesh;  $2mF_o-F_c$  (light blue mesh) map of  $ETD_B$ , ions and surrounding residues (*Right*). **C.**  $mF_o-F_c$  omit map with positive density and negative density of the exchanged ETDs shown in green and red mesh, respectively;  $2mF_o-F_c$  (light blue mesh) map of the exchanged ETDs.  $Ca^{2+}$  ion is shown in purple;  $Mg^{2+}$  ions are shown in green and water molecules are shown in red.





**Fig EV4.**

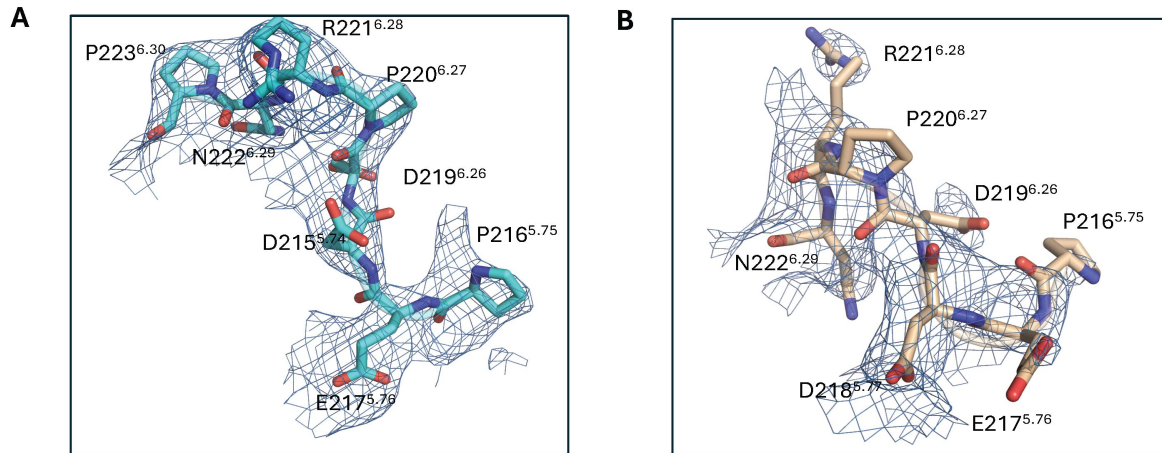
#### Electron density maps of ZLD at the active sites.

**A.**  $mF_o - F_c$  omit map with positive density of the  $ZLD_A$  shown in green mesh (*Top left*); Polder omit map of  $ZLD_A$  shown in yellow mesh;  $2mF_o - F_c$  (light blue mesh) map of  $ZLD_A$ , ions and surrounding residues (*Right*). **B.**  $mF_o - F_c$  omit map with positive density of the  $ZLD_B$  shown in green mesh (*Top left*); Polder omit map of  $ZLD_B$  shown in yellow mesh;  $2mF_o - F_c$  (blue mesh) map of  $ZLD_B$ , ions and surrounding residues (*Right*). **C.**  $mF_o - F_c$  omit map with positive density of the  $ZLD_C$  shown in green mesh (*Top left*); Polder omit map of  $ZLD_C$  shown in yellow mesh;  $2mF_o - F_c$  (blue mesh) map of  $ZLD_C$ , ions and surrounding residues (*Right*). **D.**  $mF_o - F_c$  omit map with positive density of the  $ZLD_D$  shown in green mesh (*Top left*); Polder omit map of  $ZLD_D$  shown in yellow mesh;  $2mF_o - F_c$  (blue mesh) map of  $ZLD_D$ , ions and surrounding residues (*Right*).

**Fig EV5.**

**Electron density maps of loop5-6 in the TmPPase:ETD structure.**

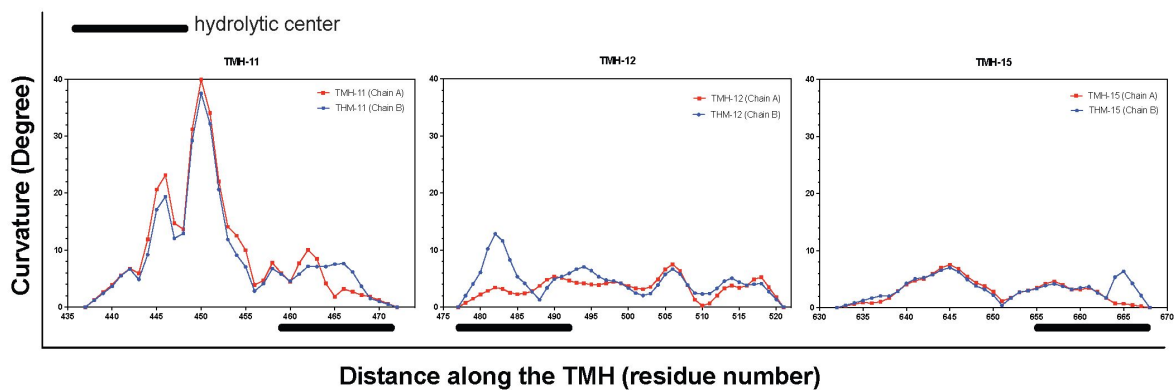
**A.**  $2mF_o - F_c$  (blue mesh) electron density map of loop5-6 at chain A. **B.**  $2mF_o - F_c$  (blue mesh) electron density map of loop5-6 at chain B.

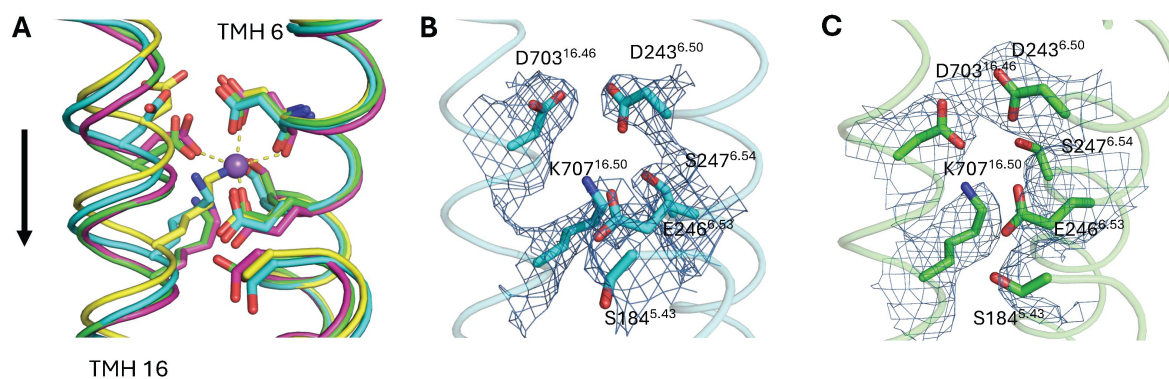


**Fig EV6.**

**Helix curvature comparison between chain A and chain B of the TmPPase:ETD structure.**

Changes in helix curvature are shown in the hydrolytic side of TMH11, TMH12, and TMH15. The black bar shows the region in the hydrolytic centre side.

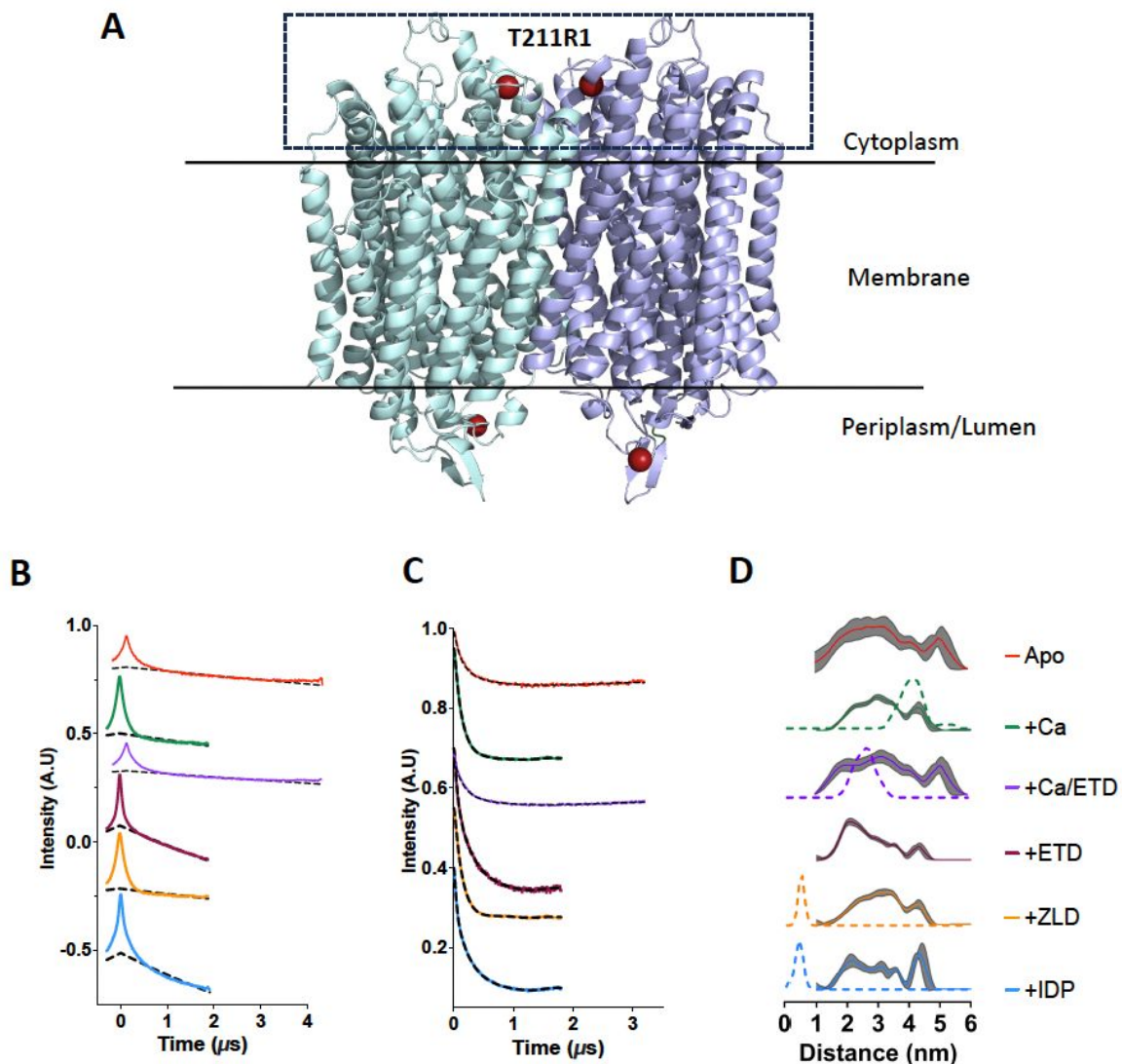




**Fig EV7.**

### **Ion gates of TmPPase structures.**

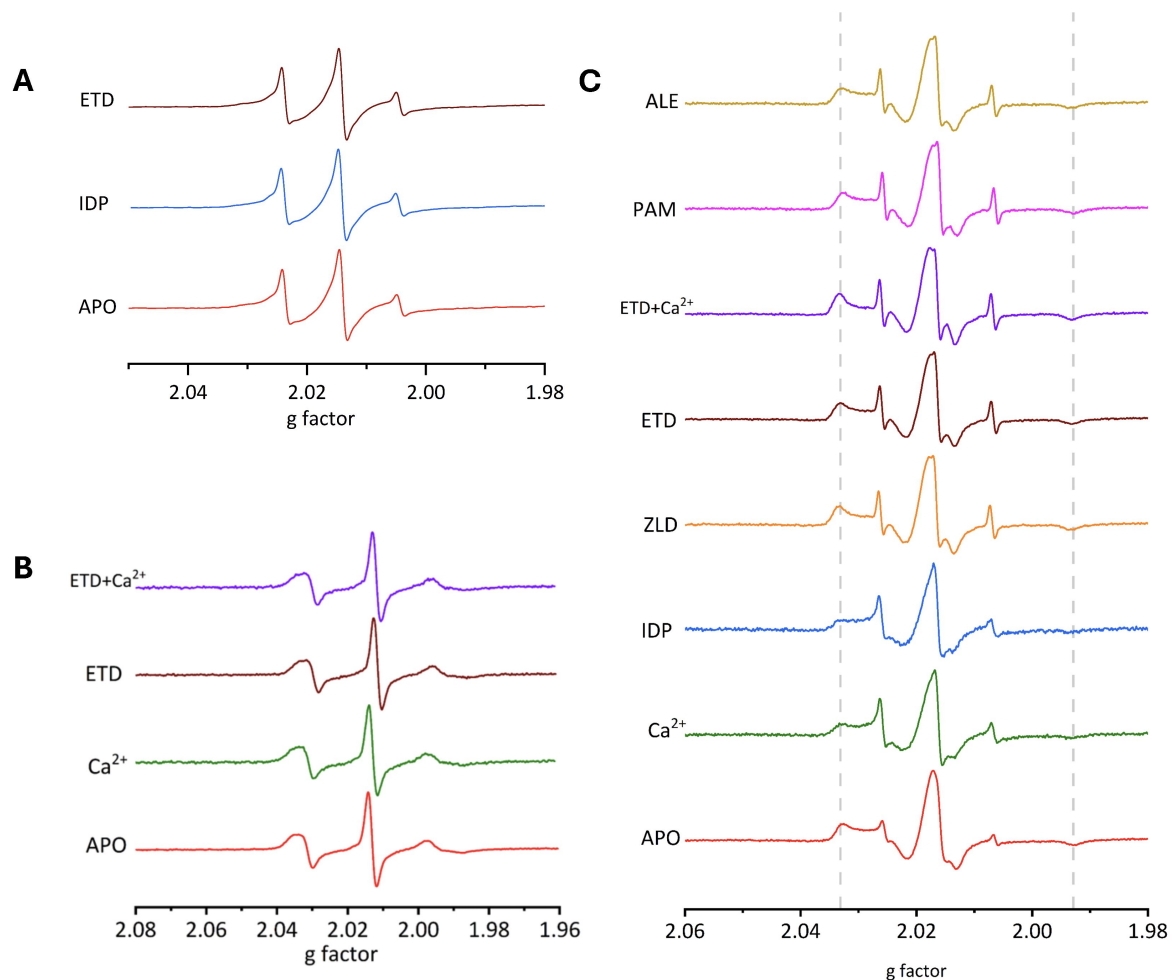
**A.** Superposition of the ion gate of the four TmPPase structures (yellow: TmPPase:Ca; cyan: TmPPase:ETD; green: TmPPase:ZLD; purple: TmPPase:IDP;  $\text{Na}^+$  is shown as a purple sphere). The movement of TMH16 is shown as the black arrow.  
**B-C.** The 2mFo-Fc and mFo-Fc density map of ion gate in the TmPPase:ETD (cyan) and TmPPase:ZLD (green) structure.



**Fig EV8.**

#### DEER distance distributions of TmPPase T211R1 under different conditions.

**A.** Structure of the TmPPase dimer (PDB 5LZQ), with monomers A and B coloured cyan and purple, respectively. The sites that were mutated to cysteine and labelled with MTSSL are shown by maroon spheres, with T211R1 on the cytoplasmic (top) side of the membrane interface. **B.** DEER raw data traces for T211R1. Each condition measured is coloured according to the condition used. **C.** DEER background-corrected time-domain traces for T211R1. **D.** The overlap between the predicted distance distribution of T211R1 from the solved crystal structures (TmPPase:Ca, TmPPase:Ca:ETD, TmPPase:ZLD, and TmPPase:IDP), shown as dashed lines, with the resulting DEER distance distributions at the respective conditions. The grey-shaded regions represent the uncertainty in the distribution. The data were all processed in DeerAnalysis2022, with validation in the same way as described in the methods.

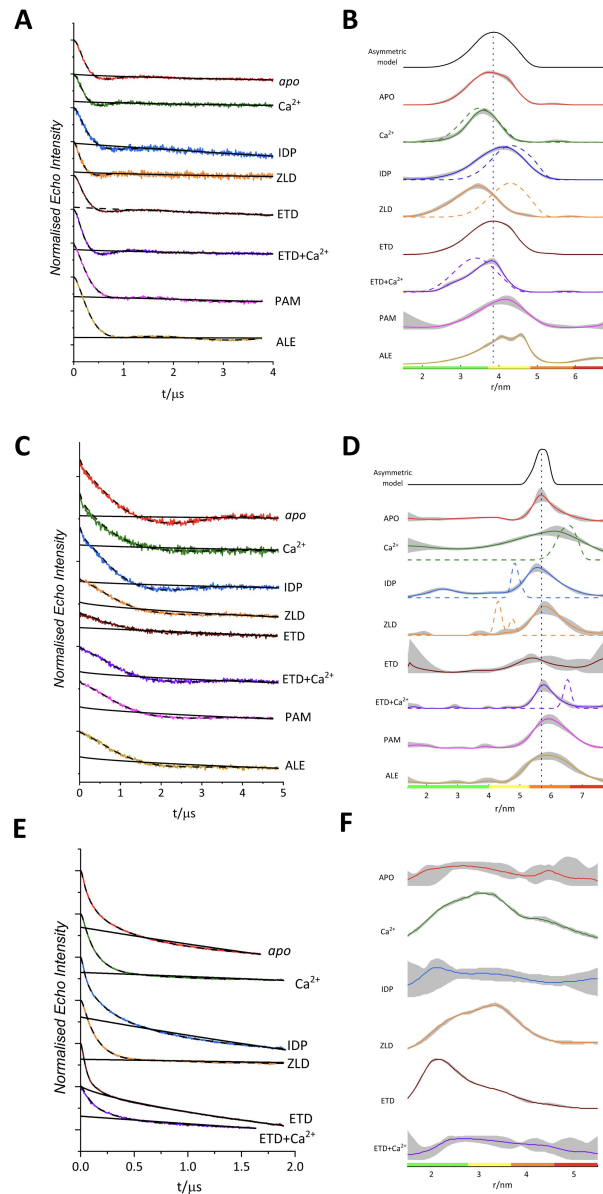


**Fig EV9.**

### CW-EPR spectra of TmPPase T211R1, C599R1, and S525R1 under different conditions

The CW-EPR data were collected at X-band frequency (9.4 GHz) and room temperature (298 K) before the addition of deuterated ethylene glycol for snap freezing. **A.** The normalised CW-EPR data for TmPPase T211R1 in its apo form (red solid line), +ETD (maroon solid line) and +IDP (cyan solid line) added. The CW-EPR spectra are vertically offset to aid visualisation. **B.** The normalised CW-EPR data for TmPPase S525R1 for apo (red solid line), +Ca (II) (green solid line), +Ca/ETD (magenta solid line) and +ETD (maroon solid line) conditions. The CW-EPR spectra are vertically offset to aid visualisation. **C.** The normalised CW-EPR data for TmPPase C599R1 for all tested conditions. The features corresponding to immobile components in the spectra are indicated by grey dashed lines. The CW-EPR spectra are vertically offset to aid visualisation.





**Fig EV10.**

### ComparativeDeerAnalyzer (CDA) data of TmPPase S525R1, C599R1 and T211R1.

**A.** The raw data for TmPPase S525R1, colour coded as in the main text. The grey solid lines correspond to three-dimensional homogeneous background functions, while the black dashed lines are the associated Tikhonov fits generated by the automated CDA software. The data are offset vertically to aid visualisation. **B.** The corresponding consensus distance distributions for TmPPase S525R1, generated by the automated CDA software. The shaded regions correspond to the 95% ( $2\sigma$ ) confidence interval. The colour scheme is consistent with panel A. **C.** The raw data for TmPPase C599R1, colour coded as in the main text. The grey solid lines correspond to three-dimensional homogeneous background functions, while the black dashed lines are the associated Tikhonov fits generated by the automated CDA software. The data are offset vertically to aid visualisation. **D.** The corresponding consensus distance distributions for TmPPase C599R1, generated by the automated CDA software. The shaded regions correspond to the 95% ( $2\sigma$ ) confidence interval. The colour scheme is consistent with panel C. **E.** The raw data for TmPPase T211R1, colour coded as in the main text. The grey solid lines correspond to three-dimensional homogeneous background functions, while the black dashed lines are the associated Tikhonov fits generated by the automated CDA software. The data are offset vertically to aid visualisation. **F.** The corresponding consensus distance distributions for TmPPase T211R1, generated by the automated CDA software. The shaded regions correspond to the 95% ( $2\sigma$ ) confidence interval. The colour scheme is consistent with panel E. The data are offset vertically to aid visualisation.

| Data Parameters           | TmPPase+Etidronate                                      | TmPPase+Zoledronate                                         |
|---------------------------|---------------------------------------------------------|-------------------------------------------------------------|
| Crystallisation condition | 0.2M CaCl <sub>2</sub> , 0.1M HEPES pH 7.0, 33% PEG 400 | 0.1M NaCl, 0.1M MES pH 6.5, 33% PEG 400, 4% ethylene glycol |
| Space group               | P 2 <sub>1</sub>                                        | P 2 <sub>1</sub> 2 <sub>1</sub> 2 <sub>1</sub>              |
| Cell dimensions           |                                                         |                                                             |
| a, b, c (Å)               | 83.7, 111.7, 105.2                                      | 101.188, 147.366, 252.341                                   |
| a, b, g (°)               | 90.0, 106.7, 90.0                                       | 90.0, 90.0, 90.0                                            |
| Source                    | DLS I03                                                 | DLS I04-1                                                   |
| Wavelength (Å)            | 0.91587                                                 | 0.97625                                                     |
| Resolution (Å)            | 80.2 - 3.15 (3.56 - 3.15)                               | 127.3 - 3.26 (3.31 - 3.26)                                  |
| Overall (Å)               | 3.15                                                    | 3.26                                                        |
| along h axis              | 3.10                                                    | 4.46                                                        |
| along k axis              | 3.60                                                    | 4.17                                                        |
| along l axis              | 4.31                                                    | 3.17                                                        |
| Measured reflections      | 130818                                                  | 445655                                                      |
| Unique reflections        | 19273                                                   | 33220                                                       |
| Completeness (%)          | 91.6 (59.3)                                             | 93.7 (72.2)                                                 |
| CC <sub>1/2</sub>         | 0.999 (0.494)                                           | 0.999 (0.573)                                               |

**Table S1.**

**X-ray data collection and refinement statistics.**

|                                             |                            |                            |
|---------------------------------------------|----------------------------|----------------------------|
| Mean I/s(I)                                 | 9.7 (1.6)                  | 10.2 (1.7)                 |
| Multiplicity                                | 6.8 (7.0)                  | 13.4 (12.2)                |
| Wilson B ( $\text{\AA}^2$ )                 | 98.7                       | 118.03                     |
| R <sub>merge</sub>                          | 0.098 (1.193)              | 0.137 (1.785)              |
| R <sub>meas</sub>                           | 0.106 (1.29)               | 0.143 (1.860)              |
| R <sub>pim</sub>                            | 0.041 (0.486)              | 0.039 (0.516)              |
| <b>Refinement</b>                           |                            |                            |
| Resolution ( $\text{\AA}$ )                 | 74.81 - 3.15 (3.27 - 3.15) | 48.22 - 3.27 (3.39 - 3.27) |
| R <sub>work</sub> (%)/R <sub>free</sub> (%) | 27.2/31.0                  | 25.9/30.4                  |
| No. of atoms                                | 10043                      | 20790                      |
| protein                                     | 10003                      | 20704                      |
| ligands                                     | 44                         | 104                        |
| water                                       | 8                          | 6                          |
| No. of chains (ASU)                         | 2                          | 4                          |
| B-factors ( $\text{\AA}^2$ )                | 88.96                      | 106.62                     |
| Protein                                     | 88.85                      | 106.53                     |
| Ligands/Ion                                 | 127.07                     | 133.77                     |
| R.m.s. deviations                           |                            |                            |
| Bond lengths ( $\text{\AA}$ )               | 0.003                      | 0.005                      |
| Bond angle ( $^\circ$ )                     | 0.63                       | 0.76                       |

**Table S1.** (continued)

**Table S1.** (continued)

|                                      |       |       |
|--------------------------------------|-------|-------|
| Ramachandran statistics <sup>†</sup> |       |       |
| Favoured (%)                         | 97.73 | 98.59 |
| Allowed (%)                          | 2.27  | 1.41  |
| Outliers (%)                         | 0.00  | 0.00  |

**Table S2.**

**Structural alignments between chains of different TmPPase structures**

| Chain                   | ETD <sub>A</sub> :ETD <sub>B</sub> | 4AV3 <sub>A</sub> :4AV3 <sub>B</sub> | ETD <sub>A</sub> :4AV3 <sub>A</sub> | ETD <sub>B</sub> :4AV3 <sub>B</sub> | ETD <sub>A</sub> :4AV3 <sub>B</sub> | ETD <sub>B</sub> :4AV3 <sub>A</sub> |
|-------------------------|------------------------------------|--------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| C <sub>α</sub> RMSD (Å) | 1.44                               | 0.39                                 | 0.72                                | 0.94                                | 0.70                                | 0.98                                |
| Chain                   | ZLD <sub>A</sub> :ZLD <sub>B</sub> | 5LZQ <sub>A</sub> :5LZQ <sub>B</sub> | ZLD <sub>A</sub> :5LZQ <sub>A</sub> | ZLD <sub>B</sub> :5LZQ <sub>B</sub> | ZLD <sub>A</sub> :5LZQ <sub>B</sub> | ZLD <sub>B</sub> :5LZQ <sub>A</sub> |
| C <sub>α</sub> RMSD (Å) | 0.51                               | 0.21                                 | 0.72                                | 0.72                                | 0.74                                | 0.69                                |
| Chain                   | ZLD <sub>A</sub> :ZLD <sub>C</sub> | ZLD <sub>A</sub> :ZLD <sub>D</sub>   | ZLD <sub>C</sub> :ZLD <sub>D</sub>  | ZLD <sub>B</sub> :ZLD <sub>D</sub>  | ZLD <sub>B</sub> :ZLD <sub>C</sub>  |                                     |
| C <sub>α</sub> RMSD (Å) | 0.58                               | 0.68                                 | 0.90                                | 0.61                                | 0.66                                |                                     |

## Acknowledgements

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## Additional information

### Author contributions

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## Additional files

**Fig EV11.** [🔗](#)



## References

- 1 Baltscheffsky H, Von Stedingk LV, Heldt HW, Klingenberg M. (1966) **Inorganic pyrophosphate: formation in bacterial photophosphorylation** *Science* **153**:1120–1122 <https://doi.org/10.1126/science.153.3740.1120> | Google Scholar
- 2 Karlsson J (1975) **Membrane-bound potassium and magnesium ion-stimulated inorganic pyrophosphatase from roots and cotyledons of sugar beet (*Beta vulgaris* L)** *Biochimica et Biophysica Acta* **399**:356–363 [https://doi.org/10.1016/0304-4165\(75\)90264-0](https://doi.org/10.1016/0304-4165(75)90264-0) | Google Scholar
- 3 Rea PA, Kim Y, Sarafian V, Poole RJ, Davies JM, Sanders D (1992) **Vacuolar H<sup>+</sup>-translocating pyrophosphatases: a new category of ion translocase** *Trends in Biochemical Sciences* **17**:348–353 [https://doi.org/10.1016/0968-0004\(92\)90313-X](https://doi.org/10.1016/0968-0004(92)90313-X) | Google Scholar
- 4 Baykov AA, Malinen AM, Luoto HH, Lahti R (2013) **Pyrophosphate-fueled Na<sup>+</sup> and H<sup>+</sup> transport in prokaryotes** *Microbiology and Molecular Biology Reviews* **77**:267–276 <https://doi.org/10.1128/mmb.00003-13> | Google Scholar
- 5 Serrano A, Pérez-Castiñeira JR, Baltscheffsky M, Baltscheffsky H (2007) **H<sup>+</sup>-PPases: yesterday, today and tomorrow** *Iubmb Life* **59**:76–83 <https://doi.org/10.1080/15216540701258132> | Google Scholar
- 6 Kajander T, Kellosalo J, Goldman A (2013) **Inorganic pyrophosphatases: one substrate, three mechanisms** *FEBS Letters* **587**:1863–1869 <https://doi.org/10.1016/j.febslet.2013.05.003> | Google Scholar
- 7 Strauss J, Wilkinson C, Vidilaseris K, de Castro Ribeiro OM, Liu J, Hillier J, Wichert M, Malinen AM, Gehl B, Jeuken LJ, Pearson AR, Goldman A. (2024) **Functional and structural asymmetry suggest a unifying principle for catalysis in membrane-bound pyrophosphatases** *EMBO Reports* **25**:853–875 <https://doi.org/10.1038/s44319-023-00037-x> | Google Scholar
- 8 Kellosalo J, Kajander T, Kogan K, Pokharel K, Goldman A (2012) **The structure and catalytic cycle of a sodium-pumping pyrophosphatase** *Science* **337**:473–476 <https://doi.org/10.1126/science.1222505> | Google Scholar
- 9 Li KM, Wilkinson C, Kellosalo J, Tsai JY, Kajander T, Jeuken LJ, Sun YJ, Goldman A (2016) **Membrane pyrophosphatases from *Thermotoga maritima* and *Vigna radiata* suggest a conserved coupling mechanism** *Nature Communications* **7**:13596 <https://doi.org/10.1038/ncomms13596> | Google Scholar
- 10 Vidilaseris K, Kiriazis A, Turku A, Khattab A, Johansson NG, Leino TO, Kiuru PS, Boije Af Gennäs G, Meri S, Yli-Kauhaluoma J, Xhaard H, Goldman A (2019b) **Asymmetry in catalysis by *Thermotoga maritima* membrane-bound pyrophosphatase demonstrated by a nonphosphorus allosteric inhibitor** *Science Advances* **5**:eaav7574 <https://doi.org/10.1126/sciadv.aav7574> | Google Scholar

- 11 Lin SM, Tsai JY, Hsiao CD, Huang YT, Chiu CL, Liu MH, Tung JY, Liu TH, Pan RL, Sun YJ (2012) **Crystal structure of a membrane-embedded H<sup>+</sup>-translocating pyrophosphatase** *Nature* **484**:399–403 <https://doi.org/10.1038/nature10963> | Google Scholar
- 12 Tsai JY, Tang KZ, Li KM, Hsu BL, Chiang YW, Goldman A, Sun YJ (2019) **Roles of the hydrophobic gate and exit channel in *Vigna radiata* pyrophosphatase ion translocation** *Journal of Molecular Biology* **431**:1619–1632 <https://doi.org/10.1016/j.jmb.2019.03.009> | Google Scholar
- 13 Tsai JY, Kellosalo J, Sun YJ, Goldman A (2014) **Proton/sodium pumping pyrophosphatases: the last of the primary ion pumps** *Current Opinion in Structural Biology* **27**:38–47 <https://doi.org/10.1016/j.sbi.2014.03.007> | Google Scholar
- 14 Ballesteros JA, Weinstein H (1995) **Integrated methods for the construction of three-dimensional models and computational probing of structure-function relations in G protein-coupled receptors** *Methods in Neurosciences* **25**:366–428 [https://doi.org/10.1016/S1043-9471\(05\)80049-7](https://doi.org/10.1016/S1043-9471(05)80049-7) | Google Scholar
- 15 Johansson NG, Dreano L, Vidilaseris K, Khattab A, Liu J, Lasbleiz A, Ribeiro O, Kiriazis A, Boije Af Gennäs G, Meri S, Goldman A, Yli-Kauhaluoma J, Xhaard H (2021) **Exploration of Pyrazolo [1, 5-a] pyrimidines as membrane-bound pyrophosphatase inhibitors** *ChemMedChem* **16**:3360–3367 <https://doi.org/10.1002/cmdc.202100392> | Google Scholar
- 16 Johansson NG, Turku A, Vidilaseris K, Dreano L, Khattab A, Ayuso Pérez D, Wilkinson A, Zhang Y, Tamminen M, Grazhdankin E, Kiriazis A, Fishwick CWG, Meri S, Yli-Kauhaluoma J, Goldman A, Boije Af Gennäs G, Xhaard H (2020) **Discovery of membrane-bound pyrophosphatase inhibitors derived from an isoxazole fragment** *ACS Medicinal Chemistry Letters* **11**:605–610 <https://doi.org/10.1021/acsmmedchemlett.9b00537> | Google Scholar
- 17 Shah NR, Vidilaseris K, Xhaard H, Goldman A (2016) **Integral membrane pyrophosphatases: a novel drug target for human pathogens?** *Aims Biophysics* **3**:171–194 <https://doi.org/10.3934/biophy.2016.1.171> | Google Scholar
- 18 Artukka E, Luoto HH, Baykov AA, Lahti R, Malinen AM (2018) **Role of the potassium/lysine cationic center in catalysis and functional asymmetry in membrane-bound pyrophosphatases** *Biochemical Journal* **475**:1141–1158 <https://doi.org/10.1042/BCJ20180071> | Google Scholar
- 19 Anashkin VA, Malinen AM, Bogachev AV, Baykov AA (2021) **Catalytic asymmetry in homodimeric H<sup>+</sup>-pumping membrane pyrophosphatase demonstrated by non-hydrolyzable pyrophosphate analogs** *International Journal of Molecular Sciences* **22**:9820 <https://doi.org/10.3390/ijms22189820> | Google Scholar
- 20 Drake MT, Clarke BL, Khosla S (2008) **Bisphosphonates: mechanism of action and role in clinical practice** *Mayo Clinic Proceedings* **83**:1032–1045 <https://doi.org/10.4065/83.9.1032> | Google Scholar
- 21 Jeschke G (2018) **The contribution of modern EPR to structural biology** *Emerging Topics in Life Sciences* **2**:9–18 <https://doi.org/10.1042/ETLS20170143> | Google Scholar
- 22 Schiemann O, Prisner TF (2007) **Long-range distance determinations in biomacromolecules by EPR spectroscopy** *Quarterly Reviews of Biophysics* **40**:1–53 <https://doi.org/10.1017/S003358350700460x> | Google Scholar

- 23 Goldfarb D (2022) **Exploring protein conformations in vitro and in cell with EPR distance measurements** *Current Opinion in Structural Biology* **75**:102398 <https://doi.org/10.1016/j.sbi.2022.102398> | [Google Scholar](#)
- 24 McHaourab HS, Steed PR, Kazmier K (2011) **Toward the fourth dimension of membrane protein structure: insight into dynamics from spin-labeling EPR spectroscopy** *Structure* **19**:1549–1561 <https://doi.org/10.1016/j.str.2011.10.009> | [Google Scholar](#)
- 25 Bordignon E, Kucher S, Polyhach Y (2019) **EPR techniques to probe insertion and conformation of spin-labeled proteins in lipid bilayers** In: Kleinschmidt JH, editors. *Lipid-Protein Interactions: Methods and Protocols* New York: Springer pp. 493–528 [Google Scholar](#)
- 26 Hartley AM, Ma Y, Lane BJ, Wang B, Pliotas C (2020) **Using pulsed EPR in the structural analysis of integral membrane proteins** In: Chechik V, Murphy DM, Bode BE, editors. *Electron Paramagnetic Resonance: Volume 27* Cambridge: The Royal Society of Chemistry pp. 74–108 [Google Scholar](#)
- 27 Pliotas C, Ward R, Branigan E, Rasmussen A, Hagelueken G, Huang H, Black SS, Booth IR, Schiemann O, Naismith JH (2012) **Conformational state of the MscS mechanosensitive channel in solution revealed by pulsed electron-electron double resonance (PELDOR) spectroscopy** *Proceedings of the National Academy of Sciences* **109**:E2675–E2682 <https://doi.org/10.1073/pnas.1202286109> | [Google Scholar](#)
- 28 Shah A, Wort JL, Ma Y, Pliotas C (2025) **Enabling structural biological electron paramagnetic resonance spectroscopy in membrane proteins through spin labelling** *Current Opinion in Chemical Biology* **84**:102564 <https://doi.org/10.1016/j.cbpa.2024.102564> | [Google Scholar](#)
- 29 Kapsalis C, Wang B, El Mkami H, Pitt SJ, Schnell JR, Smith TK, Lippiat JD, Bode BE, Pliotas C (2019) **Allosteric activation of an ion channel triggered by modification of mechanosensitive nano-pockets** *Nature Communications* **10**:4619 <https://doi.org/10.1038/s41467-019-12591-x> | [Google Scholar](#)
- 30 Kapsalis C, Ma Y, Bode BE, Pliotas C (2020) **In-lipid structure of pressure-sensitive domains hints mechanosensitive channel functional diversity** *Biophysical Journal* **119**:448–459 <https://doi.org/10.1016/j.bpj.2020.06.012> | [Google Scholar](#)
- 31 Gopinath A, Rath T, Morgner N, Joseph B (2024) **Lateral gating mechanism and plasticity of the  $\beta$ -barrel assembly machinery complex in micelles and Escherichia coli** *PNAS Nexus* **3**:pgae019 <https://doi.org/10.1093/pnasnexus/pgae019> | [Google Scholar](#)
- 32 Galazzo L, Meier G, Janulienė D, Parey K, De Vecchis D, Striednig B, Hilbi H, Schäfer LV, Kuprov I, Moeller A, Bordignon E, Seeger MA (2022) **The ABC transporter MsbA adopts the wide inward-open conformation in E. coli cells** *Science Advances* **8**:eabn6845 <https://doi.org/10.1126/sciadv.abn6845> | [Google Scholar](#)
- 33 Thaker TM, Mishra S, Zhou W, Mohan M, Tang Q, Faraldo-Goméz JD, McHaourab HS, Tomasiak TM (2022) **Asymmetric drug binding in an ATP-loaded inward-facing state of an ABC transporter** *Nature Chemical Biology* **18**:226–235 <https://doi.org/10.1038/s41589-021-00936-x> | [Google Scholar](#)

- 34 Wingler LM, Elgeti M, Hilger D, Latorraca NR, Lerch MT, Staus DP, Dror RO, Kobilka BK, Hubbell WL, Lefkowitz RJ (2019) **Angiotensin analogs with divergent bias stabilize distinct receptor conformations** *Cell* **176**:468–478 <https://doi.org/10.1016/j.cell.2018.12.005> | Google Scholar
- 35 Haysom SF, Machin J, Whitehouse JM, Horne JE, Fenn K, Ma Y, El Mkami H, Böhringer N, Schäberle TF, Ranson NA, Radford SE, Pliotas C (2023) **Darobactin B stabilises a lateral-closed conformation of the BAM complex in E. coli cells** *Angewandte Chemie International Edition* **62**:e202218783 <https://doi.org/10.1002/anie.202218783> | Google Scholar
- 36 Beck M, Covino R, Hänelt I, Müller-Mcnicoll M (2024) **Understanding the cell: Future views of structural biology** *Cell* **187**:545–562 <https://doi.org/10.1016/j.cell.2023.12.017> | Google Scholar
- 37 Lane BJ, Ma Y, Yan N, Wang B, Ackermann K, Karamanos TK, Bode BE, Pliotas C (2024) **Monitoring the conformational ensemble and lipid environment of a mechanosensitive channel under cyclodextrin-induced membrane tension** *Structure* **32**:739–750 <https://doi.org/10.1016/j.str.2024.02.020> | Google Scholar
- 38 Zhao J, Elgeti M, O’Brien ES, Sár CP, Ei Daibani A, Heng J, Sun X, White E, Che T, Hubbell WL, Kobilka BK, Chen C (2024) **Ligand efficacy modulates conformational dynamics of the  $\mu$ -opioid receptor** *Nature* **629**:474–480 <https://doi.org/10.1038/s41586-024-07295-2> | Google Scholar
- 39 Gordon-Weeks R, Parmar S, Davies TE, Leigh RA (1999) **Structural aspects of the effectiveness of bisphosphonates as competitive inhibitors of the plant vacuolar proton-pumping pyrophosphatase** *Biochemical Journal* **337**:373–377 <https://doi.org/10.1042/bj3370373> | Google Scholar
- 40 Vidilaseris K, Johansson NG, Turku A, Kiriazis A, Boije Af Gennäs G, Yli-Kauhalauma J, Xhaard H, Goldman A (2019a) **Screening for Thermotoga maritima membrane-bound pyrophosphatase inhibitors** *Journal of Visualized Experiments* **153**:e60619 <https://doi.org/10.3791/60619> | Google Scholar
- 41 Yamagata S, Iwama T (1999) **Determination of a small quantity of cystine in the presence of a large amount of cysteine** *Bioscience, Biotechnology, and Biochemistry* **63**:1503–1505 <https://doi.org/10.1271/bbb.63.1503> | Google Scholar
- 42 Tickle I, Flensburg C, Keller P, Paciorek W, Sharff A, Vonrhein C, Bricogne G (2016) **Staraniso** Cambridge, UK: Global Phasing Ltd | Google Scholar
- 43 Dahl ACE, Chavent M, Sansom MS (2012) **Bendix: intuitive helix geometry analysis and abstraction** *Bioinformatics* **28**:2193–2194 <https://doi.org/10.1093/bioinformatics/bts357> | Google Scholar
- 44 Hagelueken G, Ward R, Naismith JH, Schiemann O (2012) **MtsslWizard: in silico spin-labeling and generation of distance distributions in PyMOL** *Applied Magnetic Resonance* **42**:377–391 <https://doi.org/10.1007/s00723-012-0314-0> | Google Scholar
- 45 Tessmer MH, Stoll S (2023) **chiLife: An open-source Python package for in silico spin labeling and integrative protein modeling** *PLoS Computational Biology* **19**:e1010834 <https://doi.org/10.1371/journal.pcbi.1010834> | Google Scholar
- 46 Song Y, Dimaio F, Wang RYR, Kim D, Miles C, Brunette T, Thompson J, Baker D (2013) **High-resolution comparative modeling with RosettaCM** *Structure* **21**:1735–1742 <https://doi.org/10.1016/j.str.2013.05.005> | Google Scholar

- 47 Shah NR, Wilkinson C, Harborne SP, Turku A, Li KM, Sun YJ, Harris S, Goldman A (2017) **Insights into the mechanism of membrane pyrophosphatases by combining experiment and computer simulation** *Structural Dynamics* **4**:032105 <https://doi.org/10.1063/1.4978038> | [Google Scholar](#)
- 48 Jeschke G, Chechik V, Ionita P, Godt A, Zimmermann H, Banham J, Timmel CR, Hilger D, Jung H (2006) **DeerAnalysis2006—a comprehensive software package for analyzing pulsed ELDOR data** *Applied Magnetic Resonance* **30**:473–498 <https://doi.org/10.1007/bf03166213> | [Google Scholar](#)
- 49 Ibáñez L Fábregas, Jeschke G, Stoll S (2020) **DeerLab: A comprehensive toolbox for analyzing dipolar EPR spectroscopy data** *Magnetic Resonance Discussions* **2020**:209–224 <https://doi.org/10.5194/mr-1-209-2020> | [Google Scholar](#)
- 50 Peter MF, Gebhardt C, Mächtel R, Muñoz GGM, Glaenger J, Narducci A, Thomas GH, Cordes T, Hagelueken G (2022) **Cross-validation of distance measurements in proteins by PELDOR/DEER and single-molecule FRET** *Nature Communications* **13**:4396 <https://doi.org/10.1038/s41467-022-31945-6> | [Google Scholar](#)
- 51 Klose D, Holla A, Gmeiner C, Nettels D, Ritsch I, Bross N, Yulikov M, Allain FH, Schuler B, Jeschke G (2021) **Resolving distance variations by single-molecule FRET and EPR spectroscopy using rotamer libraries** *Biophysical Journal* **120**:4842–4858 <https://doi.org/10.1016/j.bpj.2021.09.021> | [Google Scholar](#)
- 52 Pliotas C (2017) **Ion Channel Conformation and Oligomerization Assessment by Site-Directed Spin Labeling and Pulsed-EPR** *Methods in Enzymology* **594**:203–242 <https://doi.org/10.1016/bs.mie.2017.05.013> | [Google Scholar](#)
- 53 Pliotas C, Dahl AC, Rasmussen T, Mahendran KR, Smith TK, Marius P, Gault J, Banda T, Rasmussen A, Miller S, Robinson CV, Bayley H, Sansom MS, Booth IR, Naismith JH (2015) **The role of lipids in mechanosensation** *Nature Structural & Molecular Biology* **22**:991–998 <https://doi.org/10.1038/nsmb.3120> | [Google Scholar](#)
- 54 Bhattacharyya A (1946) **On a measure of divergence between two multinomial populations** *Sankhyā: The Indian Journal of Statistics* **7**:401–406 [Google Scholar](#)
- 55 Bazzone A, Barthmes M, Fendler K (2017) **SSM-based electrophysiology for transporter research** *Methods in Enzymology* **594**:31–83 <https://doi.org/10.1016/bs.mie.2017.05.008> | [Google Scholar](#)
- 56 Malinen AM, Anashkin VA, Orlov VN, Bogachev AV, Lahti R, Baykov AA (2022) **Pre-steady-state kinetics and solvent isotope effects support the “billiard-type” transport mechanism in Na<sup>+</sup>-translocating pyrophosphatase** *Protein Science* **31**:e4394 <https://doi.org/10.1002/pro.4394> | [Google Scholar](#)
- 57 Anashkin VA, Bogachev AV, Serebryakova MV, Zavyalova EG, Bertsova YV, Baykov AA (2025) **Rapid kinetics of H<sup>+</sup> transport by membrane pyrophosphatase: Evidence for a “direct-coupling” mechanism** *Biochemical and Biophysical Research Communications* **744**:151203 <https://doi.org/10.1016/j.bbrc.2024.151203> | [Google Scholar](#)
- 58 Kellosalo J, Kajander T, Palmgren M, López-Marqués RL, Goldman A (2011) **Heterologous expression and purification of membrane-bound pyrophosphatases** *Protein Expression and*



- 59 López-Marqués RL, Pérez-Castiñeira JR, Buch-Pedersen MJ, Marco S, Rigaud JL, Palmgren MG, Serrano A (2005) **Large-scale purification of the proton pumping pyrophosphatase from *Thermotoga maritima*: A “Hot-Solve” method for isolation of recombinant thermophilic membrane proteins** *Biochimica et Biophysica Acta (BBA)-Biomembranes* **1716**:69–76 <https://doi.org/10.1016/j.bbamem.2005.08.004> | Google Scholar
- 60 Vidilaseris K, Kelloso J, Goldman A (2018) **A high-throughput method for orthophosphate determination of thermostable membrane-bound pyrophosphatase activity** *Analytical Methods* **10**:646–651 <https://doi.org/10.1039/C7AY02558K> | Google Scholar
- 61 Kabsch W (2010) **XDS** *Biological Crystallography* **66**:125–132 <https://doi.org/10.1107/S0907444909047337> | Google Scholar
- 62 McCoy AJ, Grosse-Kunstleve RW, Adams PD, Winn MD, Storoni LC, Read RJ (2007) **Phaser crystallographic software** *Applied Crystallography* **40**:658–674 <https://doi.org/10.1107/S0021889807021206> | Google Scholar
- 63 Adams PD, Afonine PV, Bunkóczi G, Chen VB, Davis IW, Echols N, Headd JJ, Hung LW, Kapral GJ, Grosse-Kunstleve RW, McCoy AJ, Moriarty NW, Oeffner R, Read RJ, Richardson DC, Richardson JS, Terwilliger TC, Zwart PH (2010) **PHENIX: a comprehensive Python-based system for macromolecular structure solution** *Biological Crystallography* **66**:213–221 <https://doi.org/10.1107/S0907444909052925> | Google Scholar
- 64 Emsley P, Lohkamp B, Scott WG, Cowtan K (2010) **Features and development of Coot** *Biological Crystallography* **66**:486–501 <https://doi.org/10.1107/S0907444910007493> | Google Scholar
- 65 Hubbell WL, Gross A, Langen R, Lietzow MA (1998) **Recent advances in site-directed spin labeling of proteins** *Current Opinion in Structural Biology* **8**:649–656 [https://doi.org/10.1016/S0959-440X\(98\)80158-9](https://doi.org/10.1016/S0959-440X(98)80158-9) | Google Scholar
- 66 Jeschke G (2012) **DEER distance measurements on proteins** *Annual Review of Physical Chemistry* **63**:419–446 <https://doi.org/10.1146/annurev-physchem-032511-143716> | Google Scholar
- 67 Lane BJ, Wang B, Ma Y, Calabrese AN, El Mkami H, Pliotas C (2022) **HDX-guided EPR spectroscopy to interrogate membrane protein dynamics** *Star Protocols* **3**:101562 <https://doi.org/10.1016/j.xpro.2022.101562> | Google Scholar
- 68 Milov A, Ponomarev A, Tsvetkov YD (1984) **Electron-electron double resonance in electron spin echo: Model biradical systems and the sensitized photolysis of decalin** *Chemical Physics Letters* **110**:67–72 [https://doi.org/10.1016/0009-2614\(84\)80148-7](https://doi.org/10.1016/0009-2614(84)80148-7) | Google Scholar
- 69 Tait CE, Stoll S (2016) **Coherent pump pulses in double electron electron resonance spectroscopy** *Physical Chemistry Chemical Physics* **18**:18470–18485 <https://doi.org/10.1039/C6CP03555H> | Google Scholar
- 70 Keller K, et al. (2016) **Averaging of nuclear modulation artefacts in RIDME experiments** *J. Magn. Reson* **272**:108–113 <https://doi.org/10.1016/j.jmr.2016.09.016> | Google Scholar
- 71 Schiemann O, Heubach CA, Abdullin D, Ackermann K, Azarkh M, Bagryanskaya EG, Drescher M, Endeward B, Freed JH, Galazzo L, Goldfarb D, Hett T, Esteban Hofer L, Fábregas Ibáñez L,



- Hustedt EJ, Kucher S, Kuprov I, Lovett JE, Meyer A, Ruthstein S, et al. (2021) **Benchmark test and guidelines for DEER/PELDOR experiments on nitroxide-labeled biomolecules** *Journal of the American Chemical Society* **143**:17875–17890 <https://doi.org/10.1021/jacs.1c07371> | [Google Scholar](#)
- 72 Russell H, Cura R, Lovett JE (2022) **DEER data analysis software: A comparative guide** *Frontiers in Molecular Biosciences* **9**:915167 <https://doi.org/10.3389/fmolb.2022.915167> | [Google Scholar](#)
- 73 Teucher M, Bordignon E (2018) **Improved signal fidelity in 4-pulse DEER with Gaussian pulses** *Journal of Magnetic Resonance* **296**:103–111 <https://doi.org/10.1016/j.jmr.2018.09.003> | [Google Scholar](#)
- 74 Chiang YW, Borbat PP, Freed JH (2005) **The determination of pair distance distributions by pulsed ESR using Tikhonov regularization** *Journal of Magnetic Resonance* **172**:279–295 <https://doi.org/10.1016/j.jmr.2004.10.012> | [Google Scholar](#)

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**Reviewer #1 (Public review):**

**Summary:**

This work examines the binding of several phosphonate compounds to a membrane-bound pyrophosphatase using several different approaches, including crystallography, electron paramagnetic resonance spectroscopy, and functional measurements of ion pumping and pyrophosphatase activity. The work synthesizes these different approaches into a model of inhibition by phosphonates in which the two subunits of the functional dimer interact differently with the phosphonate. This asymmetry in the two subunits of the dimer is consistent with past studies of this system.

**Strengths:**

This study integrates a variety of approaches, including structural biology, spectroscopic measurements of protein dynamics, and functional measurements. Overall, data analysis was thoughtful, with careful analysis of the substrate binding sites (for example calculation of POLDOR omit maps). This study agrees with previous studies that have detected functional asymmetry in the membrane PPase dimer.

### Reviewer #3 (Public review):

#### Summary:

Membrane-bound pyrophosphatases (mPPases) are homodimeric proteins that hydrolyze pyrophosphate and pump  $H^+/Na^+$  across membranes. They are an attractive drug target against protist pathogens. Non-hydrolysable  $PPi$  analogue bisphosphonates such as risedronate (RSD) and pamidronate (PMD) serve as primary drugs currently used. Bisphosphonates have a P-C-P bond, with their central carbon can accommodate up to two substituents, allowing a large compound variability. Here authors solved two TmPPase structures in complex with the bisphosphonates etidronate (ETD) and zoledronate (ZLD) and monitored their conformational ensemble using DEER spectroscopy in solution. These results reveal the inhibition mechanism by these compounds, which is crucial for developing future small-molecule inhibitors.

#### Strengths:

Authors show that seven different bisphosphonates can inhibit TmPPase with  $IC_{50}$  values in the micromolar range. Branched aliphatic and aromatic modifications showed weaker inhibition. High-resolution structures for TmPPase with ETD (3.2 Å) and ZLD (3.3 Å) are determined. These structures reveal the binding mode and shed light on the inhibition mechanism. The nature of modification on the bisphosphonate alters the conformation of the binding pocket. The conformational heterogeneity is further investigated using EPR/DEER spectroscopy under several conditions. Altogether, this provides convincing evidence for a distinct conformational equilibrium of TmPPase in solution and further supports the notion of asymmetric inhibitor binding at the active site, while maintaining a symmetric conformation at the periplasmic interface.

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### Author response:

The following is the authors' response to the original reviews

#### Public Reviews:

##### Reviewer #1 (Public review):

#### Summary:

*This work examines the binding of several phosphonate compounds to a membrane-bound pyrophosphatase using several different approaches, including crystallography, electron paramagnetic resonance spectroscopy, and functional measurements of ion pumping and pyrophosphatase activity. The work attempts to synthesize these different approaches into a model of inhibition by phosphonates in which the two subunits of the functional dimer interact differently with the phosphonate.*

#### Strengths:

*This study integrates a variety of approaches, including structural biology, spectroscopic measurements of protein dynamics, and functional measurements. Overall, data analysis was thoughtful, with careful analysis of the substrate binding sites (for example calculation of POLDOR omit maps).*

#### Weaknesses:

*Unfortunately, the protein did not crystallize with the more potent phosphonate inhibitors. Instead, structures were solved with two compounds with weak inhibitory constants >200 micromolar, which limits the molecular insight into compounds that could possibly be developed into small molecule inhibitors. Likewise, the authors choose to focus the spectroscopy experiments on these weaker binders, missing an opportunity to provide insight into the interaction between more potent binders and the protein.*

We acknowledge the reviewer concern regarding the choice of weaker inhibitors. We attempted cocrystallization with all available inhibitors, including those with higher potency. However, despite numerous efforts, these potent inhibitors yielded low-resolution crystals, making them unsuitable for detailed structural analysis. Therefore, we chose to focus on the weaker binders, as we were able to obtain high-quality crystal structures for these compounds. This allowed us to perform DEER spectroscopy and monitor conformational TmPPase state ensembles in solution with the added advantage of accurately analysing the data against structural models derived from X-ray crystallography. Using these weaker inhibitors enabled a more precise interpretation of the DEER data, thus providing reliable insights into the conformational dynamics and inhibition mechanism. As suggested by the reviewer, in the revised version, we add new DEER experiments, conditions and analysis on two of the more potent inhibitors (alendronate and pamidronate) to provide additional insight into their interactions. Furthermore, we also implemented additional DEER data on the cytoplasmic side of TmPPase; at a new site we identified (with the advantage of being an endogenous cysteine residue) and spin labelled (C599R1), given the DEER data for the previous T211R1 cytoplasmic site were difficult to interpret owing to the highly dynamic nature of this region. The new pair C599R1 yielded high-quality DEER traces and indicated more clearly than T211R1, distance distributions consistent with asymmetry across the sampled conditions. Again, as suggested by the reviewer, alendronate and pamidronate DEER measurements were also recorded for this site (cytoplasmic side; C599R1) as well as the periplasmic side (525R1).

*In general, the manuscript falls short of providing any major new insight into membrane-bound pyrophosphatases, which are a very well-studied system. Subtle changes in the structures and ensemble distance distributions suggest that the molecular conformations might change a little bit under different conditions, but this isn't a very surprising outcome. It's not clear whether these changes are functionally important, or just part of the normal experimental/protein ensemble variation.*

We respectfully disagree with the reviewer. The scale of motions particularly seen in solution (and now on a new reliable spin pair (C599R1) located on the cytoplasmic side) correspond to those seen in the full panoply of crystal structures of mPPases. Some proteins undergo very large conformational changes during catalysis – such as the rotary ATPase. This one does not, meaning that the precise motions we describe here are relevant and observed in solution for the first time. Conformational changes in the ensemble, whether large or small, represent essential protein motions which underlie key mPPase catalytic function. These dynamic transitions are extremely challenging to monitor, especially in so many conditions and our DEER spectroscopy data demonstrate the sensitivity and resolution necessary to monitor these subtle changes in equilibria, even if these are only a few Angstroms. For several of the conditions we investigated by DEER in solution, corresponding X-ray structures have been solved, with the derived distances agreeing well with the DEER distributions. This further validates the biological relevance of the structures, and reveals the complete conformational ensemble, intractable using other current approaches. Indeed, some conformational states were previously seen using serial time-resolved X-ray static structures and were consistent with asymmetry.

*The ZLD-bound crystal structure doesn't predict the DEER distances, and the conformation of Na<sup>+</sup> binding site sidechains in the ZLD structure doesn't predict whether sodium currents occur. This might suggest that the ZLD structure captures a conformation that does not recapitulate what is happening in solution/ a membrane.*

We agree with the reviewer that the ZLD-bound crystal structure does not predict the DEER distances. However, we believe this discrepancy arises from the steric bulkiness of ZLD inhibitor, which prevents the closure of the hydrolytic centre. Additionally, the absence of Na<sup>+</sup> at the ion gate in the ZLD-bound structure suggests that Na<sup>+</sup> transport does not occur, a conclusion further supported by our electrometric measurements. We agree with the reviewer; distances observed in the DEER experiments might represent a potential new conformation in solution, not captured by the static X-ray structure, thereby offering new insights into the dynamic nature of the protein under physiological conditions. This serves to emphasize the complementarity of the DEER approach to X-ray crystallography and redoubles the importance of using both techniques. Finally, the static X-ray structures have not captured the asymmetric conformations that must exist to explain half-of-the-sites reactivity, where DEER yields distance distributions, across all 16 cases tested here (two mutants with eight conditions each), that are consistent with asymmetry.

**Reviewer #2 (Public review):**

*Summary:*

*Crystallographic analysis revealed the asymmetric conformation of the dimer in the inhibitor-bound state. Based on this result, which is consistent with previous time-resolved analysis, authors verified the dynamics and distance between spin introduced label by DEER spectroscopy in solution and predicted possible patterns of asymmetric dimer.*

*Strengths:*

*Crystal structures with inhibitor bound provide detailed coordination in the binding pocket thus useful information for the mPPase field and maybe for drug development.*

*Weaknesses:*

*The distance information measured by DEER is advantageous for verifying the dynamics and structure of membrane protein in solution. However, regarding T211 data, which, as the authors themselves stated, lacks measurement precision, it is unclear for readers how confident one can judge the conclusion leading from these data for the cytoplasmic side.*

We thank the reviewer for acknowledging the advantageous use of the DEER methodology for identifying dynamic states of membrane proteins in solution. In our original manuscript, we used two sites in our analysis: S525 (periplasm) and T211 (cytoplasm), in which S525R1 yielded high-quality DEER data, while T211R1 yielded weak (or no) visible oscillations, leading to broad distributions for the several conditions tested. In the revised manuscript, we now added a third site at the cytoplasmic side (C599R1 located at TMH14), which yielded high-quality DEER data and comparable to S525R1. Both C599R1 and S525R1 spin pairs generated distance distributions for all 16 conditions (two mutants of eight conditions each) that were described well by the solution-state ensemble adopting a predominantly asymmetric conformation.

Furthermore, we have tailored our interpretation of the T211R1 DEER data, and refrain from using the data to draw conclusions about the TmPPase conformational ensemble in the



presence of different inhibitors. However, we still opted to include the T211R1 data in the SI because they confirm an important structural feature of mPPase in solution conditions; the intrinsically dynamic behaviour of the loop5-6 where T211 is located. This observation in solution is also consistent with our previous (Kellosalo et al., Science, 2012; Li et al., Nat. Commun, 2016; Vidilaseris et al., Sci. Adv., 2019; Strauss et al., EMBO Rep., 2024) and current X-ray crystallography data. To reiterate, we excluded T211R1 from any analysis relating to mPPase asymmetry and our conclusions were entirely based on the S525R1 and new C599R1 DEER data, which allowed us to monitor both sides on the membrane.

*The distance information for the luminal site, which the authors claim is more accurate, does not indicate either the possibility or the basis for why it is the ensemble of two components and not simply a structure with a shorter distance than the crystal structure.*

We thank the reviewer for pointing out this possibility and alternative interpretation of our DEER data. We now provide further analysis to show that our DEER data from both membrane sides reporters are highly consistent with (although they cannot completely exclude) asymmetry and rephrase to be inclusive of other possibilities. Importantly, this additional possibility does not affect the current interpretation of the data in our manuscript. Furthermore, we have removed Fig. 6 from the manuscript, and we now include a direct comparison of the in silico predicted distribution coming from the asymmetric hybrid structure with the 8 conditions tested, for both mutants (i.e. S525R1 and C599R1).

#### **Reviewer #3 (Public review):**

##### *Summary:*

*Membrane-bound pyrophosphatases (mPPases) are homodimeric proteins that hydrolyze pyrophosphate and pump H<sup>+</sup>/Na<sup>+</sup> across membranes. They are attractive drug targets against protist pathogens. Non-hydrolysable PPI analogue bisphosphonates such as risedronate (RSD) and pamidronate (PMD) serve as primary drugs currently used. Bisphosphonates have a P-C-P bond, with its central carbon can accommodate up to two substituents, allowing a large compound variability. Here the authors solved two TmPPase structures in complex with the bisphosphonates etidronate (ETD) and zoledronate (ZLD) and monitored their conformational ensemble using DEER spectroscopy in solution. These results reveal the inhibition mechanism of these compounds, which is crucial for developing future small molecule inhibitors.*

##### *Strengths:*

*The authors show that seven different bisphosphonates can inhibit TmPPase with IC<sub>50</sub> values in the micromolar range. Branched aliphatic and aromatic modifications showed weaker inhibition.*

*High-resolution structures for TmPPase with ETD (3.2 Å) and ZLD (3.3 Å) are determined. These structures reveal the binding mode and shed light on the inhibition mechanism. The nature of modification on the bisphosphonate alters the conformation of the binding pocket.*

*The conformational heterogeneity is further investigated using DEER spectroscopy under several conditions.*

##### *Weaknesses:*

*The authors observed asymmetry in the TmPPase-ELD structure above the hydrolytic center. The structural asymmetry arises due to differences in the orientation of ETD within each monomer at the active site. As a result, loop5-6 of the two monomers is oriented differently, resulting in the observed asymmetry. The authors attempt to further*

*establish this asymmetry using DEER spectroscopy experiments. However, the (over)interpretation of these data leads to more confusion than any further understanding. DEER data suggest that the asymmetry observed in the TmPPase-ELD structure in this region might be funneled from the broad conformational space under the crystallization conditions.*

We respectfully disagree with the reviewer. The asymmetry was previously established using serial time crystallography (Strauss et al., EMBO Rep, 2024) and biochemical assays (e.g. Malinen et al., Prot. Sci., 2022; Artukka et al., Biochem J, 2018; Luoto et al., PNAS, 2013) and partially seen in one static structure (Vidilaseris et al., Sci Adv 2019). DEER data here also show that the previously proposed asymmetry is also present (and this presence of asymmetry is consistent across all DEER data) within the TmPPase conformational ensemble in solution conditions. Although we cannot rule out the possibility that the TmPPase monomers adopt a metastable intermediate state, in such a case we would expect the distance changes reported by DEER to be symmetric across both membrane sides. However, we observe a symmetry breaking between the cytoplasmic and periplasmic TmPPase sites. Indeed, DEER data yield distance distributions similar to that of the hybrid asymmetric structure under all: apo, +Ca, +Ca/ETD, +ETD, +ZLD, +IDP, +PAM, +ALE conditions.

*DEER data for position T211R1 at the enzyme entrance reveal a highly flexible conformation of loop56 (and do not provide any direct evidence for asymmetry, Figure EV8).*

Please see relevant response above. We acknowledge that T211 is indeed situated on a highly dynamic loop, which is important for gating and our DEER data confirm the high flexibility of this protein region. Given we have not observed dipolar oscillations, leading to broad distributions, we have stated in the original manuscript that we will not establish the presence of any asymmetry in solution on the basis of T211, rather relying on the S525R1 and the new C599R1 sites, for which we have acquired high-quality DEER data, as was also pointed out and has been commented on by all reviewers. We have provided data at the C599R1 position (same cytoplasmic side as 211 for which we have now limited our analysis to a minimum) which further provides evidence for asymmetry, including two new conditions.

*Similarly, data for position S521R1 near the exit channel do not directly support the proposed asymmetry for ETD.*

The reviewer appears to suggest that we hold the S525R1 DEER data as direct proof of asymmetry; this is combative on the grounds that to directly prove asymmetry would require time-resolved DEER measurements, far beyond the scope of this work. Rather, we have applied DEER measurements to explore whether asymmetry (observed previously via time-resolved X-ray crystallography) is also present (or indeed a possibility) in solution. All our S525R1 and C599R1 DEER data (recorded for eight conditions) are consistent with asymmetry (see also detailed response above).

*Despite the high quality of the data, they reveal a very similar distance distribution. The reported changes in distances are very small ( $\pm 0.3$  nm), which can be accommodated by a change of spin label rotamer distribution alone. Further, these spin labels are located on a flexible loop, thereby making it difficult to directly relate any distance changes to the global conformation*

We thank the reviewer for recognising the high quality of our DEER data for the S525R1 site which we now complement with a new pair on the cytoplasmic facing membrane side (C599R1) with DEER data of comparable quality as for S525R1, where visual oscillations in the raw traces for both spin pairs, as in our case, reportedly lead to highly accurate and reliable

distributions, able to separate (in fortuitous cases) helical movements of only a few Angstroms (Peter et al., Nature Comms 13:4396, 2022; Klose et al., Biophys J 120:4842-4858, 2021). The ability of DEER/PELDOR offering near Angstrom resolution was also previously demonstrated by the acquisition and solution of highresolution multi-subunit spin-labelled membrane protein structures (Pliotas et al., PNAS, 2012; Pliotas et al., Nat Struct Mol Biol, 2015; Pliotas, Methods Enzymol, 2017) as well as its ability in detecting small (and of similar to mPPase magnitude) conformational changes in different integral membrane protein systems (Kapsalis et al., Nature Comms, 2019; Kubatova et al., PNAS, 2023; Schmidt et al., JACS, 2024; Lane et al., Structure, 2024; Hett et al., JACS, 2021; Zhao et al., Nature, 2024), occurring under different conditions and/or stimuli in solution and/or lipid environment. The changes here are not below the detection sensitivity of DEER (e.g.  $\sim 7$  Angstroms between the two modal distance extremes (+Ca vs +IDP for S525R1), and with all other conditions showing intermediate changes.

We agree with the reviewer that these changes are relatively small, but they are expected for membrane ion pumps. Indeed, none of the mPPase structures show helical movements of greater than half a turn, and that only in helices 6 and 12. There appear to be larger-scale loop closing motions of the 5-6 loop that includes T211, due to the presence of E217 which binds to one of the  $Mg^{2+}$  ions that coordinate the leaving group phosphate. This is, inter alia, the reason that this loop is so flexible: it cannot order before substrate is bound.

The reviewer suggests that the subtle distance shifts detected arise only from changes of label rotamer distribution. However, the concerted nature of the modal distance shifts with respect to multiple different conditions at a single labelling site strongly suggests that preferential rotamer orientations are not the cause. Indeed, for so many spin labels to undergo an arbitrary shift that the modal distance of the entire distribution changes – and in the absence of any conformational change – appears improbable. Here we have the resolution to detect such subtle differences by DEER, given there are unambiguous shifts in our time domain data (i.e. the position of the minimum of the first dipolar oscillation) (Fig 4) and these are reflected in the modal distances in the distributions. We also refrain from performing any quantitative analysis and use qualitative trends in modal distance shifts only; all which support our proposed model of a symmetry breaking across the membrane face. To further belabour this point, we do not quantify the DEER data (for instance through parametric fitting) to extract populations of different conformational states and we appreciate that to do so would be highly prone to error; however we do (and can, we feel without over-interpretation) assert that the modal distances shift.

*The interpretations listed below are not supported by the data presented:*

*(1) 'In the presence of  $Ca^{2+}$ , the distance distribution shifts towards shorter distances, suggesting that the two monomers come closer at the periplasmic side, and consistent with the predicted distances derived from the TmPPase:Ca structure.'*

*Problem: This is a far-stretched interpretation of a tiny change, which is not reliable for the reasons described in the paragraph above.*

While the authors overall agree with the reviewer assessment that  $\pm 0.3$  nm is a small (not a minor) change, there are literature examples quantifying (or using for quantification) distribution peaks separated by similar  $\Delta r$ . (Kubatova et al., PNAS, 2023; Schmidt et al., JACS, 2024; Hett et al., JACS, 2021; Zhao et al., Nature, 2024). However, the time-domain data clearly indicate the position of the first minimum of the dipolar oscillation shifts to shorter dipolar evolution time. The sensitivity of the time-domain data to subtle changes in dipolar coupling frequency is significantly improved compared to the distance distributions.

Importantly, we have fitted Gaussians to the experimental distance distributions of 525R1 output by the Comparative Deer Analyzer 2.0 and observed a change in the distribution width

in presence of  $\text{Ca}^{2+}$ , implying the rotameric freedom of the spin label is restricted. However, the CW-EPR for S525R1 indicate that the rotational correlation time of the spin label is highly consistent between conditions (the spectra are almost identical); this cannot be explained simply by rotameric preference of the spin label (as asserted by the reviewer 3), as there is no (further) immobilisation observed from the CW-EPR of apo-state (Figure EV9) to that in presence of  $\text{Ca}^{2+}$ . Furthermore, in the absence of conformational changes, it is reasonable to assume (and demonstrable from the CW-EPR data) that the rotamer cloud should not significantly change between conditions. However, Gaussian fits of the two extreme cases yielding the longest (i.e., in presence of IDP) and shortest (in presence of ZLD) modal distances for the S525R1 DEER data indicated significant (i.e., above the noise floor after Tikhonov validation) probability density for the IDP condition at 50 Å ( $P(r) = 0.18$ ). This occurs at four standard deviations above the mean of the Gaussian fit to the +ZLD condition, which by random chance should occur with <0.007% probability.

As in previous response, the method can detect changes of such magnitude which are not small, but physiologically relevant and expected for integral membrane proteins, such as mPPases. Indeed, even in equal (or more) complex systems such as heptameric mechanosensitive channel proteins DEER provided sub-Angstrom accuracy, when a spin labelled high resolution XRC structure was solved (Pliotas et al., PNAS, 2012; Pliotas et al., Nat Struct Mol Biol, 2015). Despite this being an ideal case where DEER accuracy was experimentally validated another high-resolution structural method on modified membrane protein and is not very common it demonstrates the power of the method, especially when strong oscillations are present in the raw DEER data (as here for mPPase S525R1, and C599R1), even when multiple distances are present, Angstrom resolution is achievable in such challenging protein classes.

*(2) 'Based on the DEER data on the IDP-bound TmPPase, we observed significant deviations between the experimental and the in silico distances derived from the TmPPase:IDP X-ray structure for both cytoplasmic- (T211R1) and periplasmic-end (S525R1) sites (Figure 4D and Figure EV8D). This deviation could be explained by the dimer adopting an asymmetric conformation under the physiological conditions used for DEER, with one monomer in a closed state and the other in an open state.'*

*Problem: The authors are trying to establish asymmetry using the DEER data. Unfortunately, no significant difference is observed (between simulation and experiment) for position 525 as the authors claim (Figure 4D bottom panel). The observed difference for position 112 must be accounted for by the flexibility and the data provide no direct evidence for any asymmetry.*

Reviewer 3 is incorrect in suggesting that we are trying to prove asymmetry through the DEER data. That is a well-known fact in the literature (e.g. Vidilaseris et al, Sci Adv 2019) where we show (1) that the exit channel inhibitor ATC (i.e. close to S525R1) binds better in solution to the TmPPase:PPi complex than the TmPPase:PPi<sub>2</sub> complex, and (2) that ATC binds in an asymmetric fashion to the TmPPase:IDP<sub>2</sub> complex with just one ATC dimer on one of the exit channels. We merely use the DEER data to support this well-established fact.

However, because we agree that the DEER data in presence of IDP does not provide direct proof for asymmetry; particularly for the cytoplasmic facing mutant T211R1, we have refrained from interpreting T211R1 data beyond being a highly dynamic loop region (as evidenced by the broad distributions). As pointed out by the reviewer, the differences in distance distributions between conditions observed for T211R1 likely arise from conformational heterogeneity in solution. Furthermore, we now report DEER data on another new site (C599R1), which is also on the cytoplasmic side and yields high quality DEER data comparable to the S525R1 data (commended for their quality by both the reviewers). The C599R1 measurements show that in all conditions tested, highly similar distributions are

observed, inconsistent with the *in silico* predicted distance distributions from the symmetric X-ray structures, but consistent with an asymmetric hybrid structure (i.e. open-closed) in solution. Importantly, the difference between the fully open (6.8 nm modal distance) and fully closed (4.8 nm modal distance) states of the C599R1 dimer is larger than for the S525R1 dimer pair. Thus, delineating the asymmetric hybrid conformation from the symmetric conformations is more robust.

(3) *'Our new structures, together with DEER distance measurements that monitor the conformational ensemble equilibrium of TmPPase in solution, provide further solid experimental evidence of asymmetry in gating and transitional changes upon substrate/inhibitor binding.'*

*Problem: See above. The DEER data do not support any asymmetry.*

We feel that the reviewer comments here are somewhat unfounded. All the DEER data (for 525R1 periplasmic and C599R1 cytoplasmic sites are described, most parsimoniously, using an asymmetric hybrid structure. In particular, the new C599R1 distance distributions are poorly described by the symmetric X-ray crystal structures, with a conserved modal distance of approx. 5.8 nm throughout the tested conditions that aligns nicely with the *in silico* predictions from the asymmetric hybrid structure. Additionally, all S525R1 and C599R1 data well exceed the relevant criteria of the recent white paper (Schiemann et al., 2021, JACS) from the EPR community to be considered reliably interpretable (strong visual oscillations in the raw traces; signal-to-noise ratio .r.t modulation depth of > 20 in all cases; replicates have been performed and added into the maintext or supplementary; near quantitative labelling efficiency (evidenced by lack of free spin label signal in the CW-EPR spectra); analysed using the CDA (now Figure EV10) to avoid confirmation bias).

While the DEER data do not prove asymmetry, we do not claim proof of asymmetry in the above sentence. We concede to rephrase the offending sentence above as: “Our new structures, together with DEER distance measurements that monitor the conformational ensemble of TmPPase in solution, do not exclude asymmetry in gating and transitional changes upon substrate/inhibitor binding and are consistent with our proposed model.” We feel that this reframed conjecture of asymmetry is well founded; indeed, comparing all the 16 experimentally derived DEER distance distributions for the 525R1 and 599R1 sites with *in-silico* modelling performed on the hybridised asymmetric structure (i.e., comprised of one monomer bound to Ca<sup>2+</sup> and another bound to IDP) yields overlap coefficients (Islam and Roux, JPC B, 2015) of >0.85. This implies the envelope of the modelled distance distribution is quantitatively inside the envelope of the experimental distance distributions. Thus, the DEER data support asymmetry (previously observed by time-resolved XRC) in solution, and while we appreciate that ideally one would measure time-resolved DEER to directly correlate kinetics of conformational changes within the ensemble to the catalytic cycle of mPPase, (and this is something we aim to do in the future), it is far beyond the scope of this study.

Indeed, half-of-the-sites reactivity has been demonstrated in at least the following papers

(Vidilaseris et al, Sci Acv, 2019, Strauss et al, EMBO Rep. 2024, Malinen et al Prot Sci, 2022, Artukka et al Biochem J, 2018; Luoto et al, PNAS, 2013). Half-of-the sites activity requires asymmetry in the mechanism, and therefore asymmetric motions in the active site (viz 211) and exit channel (viz 525). As mentioned above, we have demonstrated this for other inhibitors (Vidilaseris et al 2019) and as part of a time-resolved experiment (Strauss et al 2024). In fact, given the wealth of evidence showing that the symmetrical crystal structures sample a non- or less-productive conformation of the protein, it would be quixotic to propose the DEER experiments - in solution - do not generate asymmetric conformations. It certainly doesn't obey Occam's razor of choosing the simplest possible explanation that covers the data.



(4) Based on these observations, and the DEER data for +IDP, which is consistent with an asymmetric conformation of TmPPase being present in solution, we propose five distinct models of TmPPase (Figure 7).

*Problem: Again, the DEER data do not support any asymmetry and the authors may revisit the proposed models.*

We have redressed the proposed models and limited them to four asymmetric models to clearly illustrate the *apo*+Ca/+Ca:ETD-state (model 1) and highlight the distinct binding patterns of various inhibitors (ETD, ZLD and IDP; model 2-4), which result in a variety of closed/open-open states. In this version, we clarify that the proposed models are not solely based on the DEER data but all DEER data recorded for multiple conditions, inhibitors and for two opposite membrane side facing reporters are highly consistent, and are grounded in both current and previously solved structures, with the DEER data providing additional consistency with these models.

(5) 'In model 2 (Figure 7), one active site is semi-closed, while the other remains open. This is supported by the distance distributions for S525R1 and T211R1 for +Ca/ETD informed by DEER, which agrees with the *in silico* distance predictions generated by the asymmetric TmPPase:ETD X-ray structure'

*Problem: Neither convincing nor supported by the data*

We respectfully disagree with the reviewer. However, owing to the conformational heterogeneity of T211R1, we now exclude T211R1 data from quantitative interpretation of changes to the conformational ensemble. Instead, we include new DEER data from site C599R1, which provides high-quality and convincing data that is consistent with asymmetry at the cytoplasmic face, and inconsistent with *in silico* distance distributions derived from symmetric X-ray crystal structures. Furthermore, the S525R1 distance distributions for the +ETD (corresponding to +Ca/ETD) and +ZLD conditions were directly compared with both the *apo*-state distance distribution (corresponding to a fully open, symmetric conformation) and the *in silico* predicted distributions of the asymmetric hybrid structure (corresponding to an open-closed conformation). Overlap coefficients were calculated (given in the main text) that indicated the +ETD (corresponding to +Ca/ETD) and +ZLD S525R1 distributions were more consistent with the *apo*-state distance distribution. This suggests that while on the cytosolic face of the membrane, an open-closed conformation is favoured, on the periplasmic face, a symmetric open-open conformation is favoured.

#### **Recommendations for the authors:**

##### **Reviewer #1 (Recommendations for the authors):**

(1) The DEER experiments were performed with the two crystallized inhibitors, ETD and ZLD, along with previously characterized IDP. It would increase the impact of a tighter-binding phosphonate was examined since the inhibitory mechanism of these molecules is of greater interest.

We acknowledge the reviewer concern regarding the choice of weaker inhibitors. We chose to focus on the weaker binders, as we were able to obtain high-quality crystal structures for these compounds. This allowed us to perform DEER spectroscopy with the added advantage of accurately analysing the data against structural models derived from X-ray crystallography. In the revised version, we also include results from alendronate and pamidronate, two of the tighter inhibitors, which show similar and consistent results to the others.



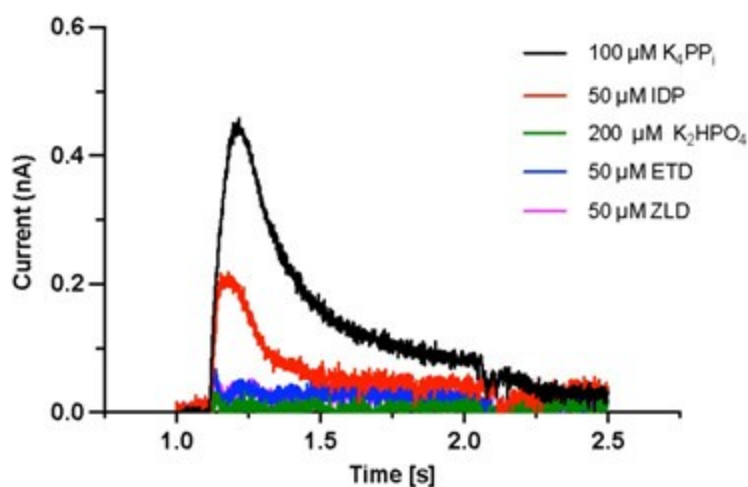
(2) I'm not able to find the concentrations of ETD and ZLD used for the DEER experiments. This information should be added to the Methods section on sample prep for EPR.

The information is already mentioned in the Method section on sample preparation for EPR spectroscopy (page 24), where we indicated that the protein aliquots were incubated with a final concentration of 2 mM inhibitors or 10 mM CaCl<sub>2</sub> (30 min, RT). However, we recognise that this may not have been sufficiently clear. To clarify, we now explicitly state that the concentration of ETD and ZLD (amongst other inhibitors) used for the DEER experiments is 2 mM.

(3) There should be additional detail about the electrometry replicates. Does "triplicate" mean three measurements on the same sensor, three different sensors, and different protein preparations? At a minimum, data should be collected from three different sensors to ensure that the negative results (lack of current) for ETD and ZLD are not due to a failed sensor prep. In addition, Data from the other replicates should be shown in a supplementary figure, either the traces, or in a summary figure. Are the traces shown collected on the same sensor? They could be, in principle, since the inhibitor is washed away after each perfusion.

Yes, by 'triplicate', we mean three measurements taken on the same sensor. All traces shown were collected from a single sensor. Thank you for your advice; we now show here additional data from other sensors that display the same pattern. As for the possibility of a failed sensor preparation, this is unlikely since we always ensure the sensor quality with the substrate (PPi) as a positive control after each measurement.

#### Author response image 1.



(4) I'm confused by the NEM modification assay, and I don't think there is enough information in this manuscript for a reader to figure out what is happening. Why is the protein active if an inhibitor is present? I understand that there is a conformational change in the presence of the inhibitor that buries a cysteine, but the inhibitor itself should diminish function, correct? Is the inhibitor removed before testing the function? In addition, it would be clearer if the cysteines that are modified are indicated in the main text. I don't understand what is being shown in Figure Ev2. Shouldn't the accessible cysteines in the apo form be shown? Finally, the sentence "IDP has been reported to

*prevent the NEM modification..." does not make sense to me. Should the word "by" be removed from this sentence?*

We apologize for the confusion. Yes, the inhibitors were removed before testing the protein function. In Figure EV2, the accessible cysteines are shown for both the apo and IDP-bound states. As seen, the accessible cysteines in the IDP-bound states are fewer than those in the apo state, meaning fewer cysteines are available for modification. Consequently, more activity is retained when IDP binds due to the reduction in accessible cysteines. We have addressed this in the manuscript (see the method section on the NEM modification assay).

*(5) Why does the model in Figure 7 show the small molecules bound to only one subunit, when they are crystallized in both subunits?*

We propose that the small molecules bound to the two subunits in the crystal structure is likely a result of substrate inhibition, given the excess inhibitor used during crystallisation (e.g. Artukka, et al., Biochemical Journal, 2018; Vidilaseris, et al., Science Advances, 2022). Our PELDOR data indicate that in solution, the small molecules bound to TmPPase are in an intermediate state between both subunits being closed and both being open, most likely with at least one subunit in an open state. This is also consistent with previous kinetic studies (Anashkin, V. A., et al., International Journal of Molecular Sciences, 22, 2021), which showed that the binding constant of IDP to the second subunit is around 120 times higher than that of the first subunit.

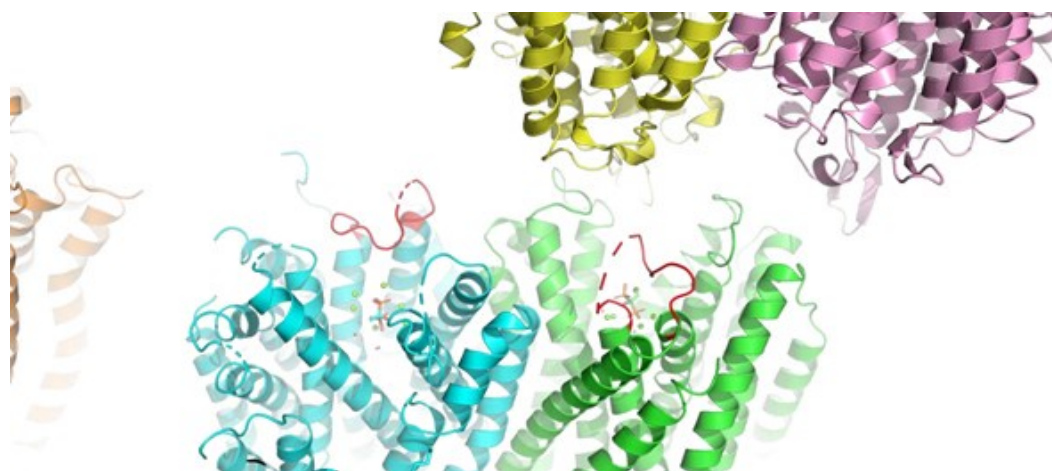
*(6) The authors argue that the two ETDs bound in the two protomers adopt distinct conformations. Can this be further supported, for example, by swapping the position of the two ETDs between the two protomers and calculating a difference map (there should be corresponding negative/positive density if the modelling of the two different conformations is robust)?*

As per the reviewer suggestion, we swapped the positions of the two ETDs between the protomers and calculated the difference electron density map. This analysis, presented in Figure EV3, reveals corresponding negative and positive electron density peaks, indicating that the ETDs indeed adopt distinct conformations in each protomer, supporting the accuracy of our modeling.

*(7) Are the changes in loop conformation possibly due to crystal packing differences for the two protomers?*

We examined the crystal packing of the two protomers and found no interactions at the loop regions (red coloured in Author response image 2 below) that could be attributed to crystal packing differences. Therefore, we rule out this possibility.

**Author response image 2.**



(8) Typos:

*Legend for Figure EV2 cystine - cysteine*

*Page 14, last sentence of the first paragraph: further - further*

*Figure 6 legend: there is no reference to panel B.*

Thanks for pointing out the typos, now they are fixed.

**Reviewer #2 (Recommendations for the authors):**

*(1) T211 is located on the same loop where ligand/inhibitor-coordinating side chains (E217, D218) are located. It has not been tested whether spin labeling here would affect inhibitor binding.*

We test all the mutant(s) activity before spin labelling, but not the activity of the spin-labelled mutants. MTSSL spin labels are typically not structurally perturbing. In particular, the T211R1 site that the reviewer is referring to is now not included in our interpretation of conformational changes occurring during mPPase's functional cycle.

*(2) Why should the spin label be introduced to T211, which is recognized as a flexible region in the crystal structure? Authors should search for suitable residues except for T211 and other residues in this loop to evaluate the cytoplasmic distance.*

We acknowledge the reviewer's concern regarding the flexibility of the T211 region for spin labelling. Given the challenges associated with TmPPase, including reduced protein expression, loss of function, or inaccessibility upon spin labelling at certain sites, we have explored alternative residues. After extensive testing, we identified C599 as a suitable site for spin labelling resulting in high-quality DEER data. The results from spin labelling at C599 have been incorporated into the revised manuscript.

*(3) On the other hand, DEER data for S525 is solid, as the authors stated. This residue is located on the luminal side of the enzyme. However, the description of the luminal side structure and the comparison of symmetric/asymmetric dimer in this par are missing in the paper.*

We thank the viewer for their positive assessment of the S525R1 DEER data. The data for 525 and now also for 599 spin pairs are indeed solid given the strong visual oscillation we observed particularly in such a challenging system.

We presented the periplasmic sites in the crystal structure dimer (Figure 4A), highlighting both the symmetrical region and the asymmetric model in Figure 4. In the revised version, we include additional details about this region and our rationale for labeling at position S525.

*(4) The conclusion models (Figure 7) are misleading. In the crystal structure, the 5-6 Loop distance between each monomer should be close given the location of the dimer interface, and the actual distance between T211 in the structure (for example, in 5lzc) is about 10Å. Nevertheless, the model depicts this distance longer than S525 (40.7Å in 5LZQ), which would give a false impression.*

We would like to apologize for the misleading model. We have now corrected the models to ensure they are consistent with their respective regions in the crystal structures.

*(5) P8 last paragraph*

*It is hard to imagine that in a crystal lattice, the straight inhibitor always binds to monomer A, and the neighboring monomer is always attached to a slightly tilted inhibitor, which causes asymmetry. For example, wouldn't it mean that it would first bind to one of them, which would then affect the neighboring monomer via 5-6 Loop, which would then affect its binding pose? So in this case, the inhibitor did not ARAISE asymmetry, and this is where it is misleading for readers.*

We apologize for the confusion. What we intended to convey is that the first inhibitor binds to one protomer, which then affects the conformation of the neighbouring monomer, ultimately influencing its binding pose. This is required for half-of-the-sites reactivity, which is well-established in this system. This is reflected in our crystal structure, where we observed asymmetry in the loop 5-6 region and the ETD orientation between the two protomers. We have addressed this in the manuscript accordingly.

*(6) P11 L4 EV10 instead of EV8?*

Thanks for pointing out. We have corrected it accordingly.

*(7) P11 L5 It is difficult to determine whether the peak is broad or sharp. Should be evaluated quantitatively by showing the half-value width of the peak. This may also be helpful to judge whether the peak is a mixture of two components or a single one.*

We have taken this analysis out and rephrased the offending sentence. We have also added the FWHM values as the Reviewer suggested, and corresponding standard deviations for the distance distributions (under approximation as Gaussian distribution).

*(8) Throughout the paper, the topology of the enzyme may be difficult to follow for readers who are not experts in this field. Please indicate the membrane plane's location or a figure's viewpoint in the caption.*

We acknowledge the importance of making our figures accessible to all readers. In the revised manuscript, we have enhanced the clarity of our figures by explicitly indicating the membrane plane's location and specifying the viewpoint in each figure caption. For example, we have added annotations such as "Top view of the superposition of chain A (cyan) and

chain B (wheat), showing the relative movements (black arrow) of helices. The membrane plane is indicated by dashed lines.”

(9) Figure 2B Check the color of the helix.

*IDP and ETD are almost the same color, so it is difficult to see the superposition. It would be easier to understand the reading by, for example, using a lighter or transparent color set only for IDPs.*

We acknowledge the reviewer concern regarding the colour similarity between the IDP and ETD in Figure 2B, which hinders clear differentiation. To enhance visual distinction, we have adjusted the colour scheme by changing the TmPPase:IDP structure colour to light blue. This modification improves the clarity of the superposition, making the structural differences more discernible.

(10) Figure 2C Check the coordination state (dotted line), there appears to be coordination between E217Cg and Mg. Also, water that is located near N492 appears to be a bit distant from Mg, why does this act as a ligand? Stereo view or view from different angles, and distance information would help the reader understand the bonding state in more detail.

Yes, we confirm that  $Mg^{2+}$  is coordinated by the oxygen atoms from both the side chain and main chain of residue E217. The water molecule near N492 is not directly coordinated with  $Mg^{2+}$  but interacts with the O5 atom of one of the phosphate groups in ETD. To enhance clarity, we have updated Figure 2C (and other related figures) to include stereo views.

(11) Figure 5A: in the Bottom view (lower left), the symmetric dimer does not look symmetric. Better to view from a 2-fold axis exactly.

We have taken this figure out entirely and instead add a direct comparison to the in silico predicted distribution from the asymmetric hybrid structure to all 16 experimental DEER distributions. We have added the symmetric and asymmetric structures to Fig. 4A and view the symmetric structure along the 2-fold axis, as suggested.

(12) Figure 5B: Indicate which data is plotted in the caption.

As mentioned above, we have taken this figure out, as we felt quantifying two overlapping populations from a single Gaussian was over-interpretation of the data, and at the suggestion of reviewer 3, we have tailored our interpretation here.

(13) Figure EV8:

*Because the authors discuss a lot about their conclusive model based on this data, Figure EV8 should be treated as a main figure, not a supplement. However, this reviewer has serious concerns about the measurement in this figure. Because DEER for T211 is too noisy, I don't see the point in discussing this in detail. For example, in the Ca/ETD data, there is a peak near 50A, but it would be difficult for TM5 to move away from this distance unless the protein unfolds. I do not find it meaningful to discuss using measurement results in which such an impossible distance is detected as a peak.*

*A: Show top view as in Figure 5*

*D: 2nd row dotted line. Regarding the in silico model that is used as a reference to compare the distance information, the distance of 40-50 A for T211 in the Ca-bound form is hard to imagine. PDB 4av6 model shows that T211 is disordered and not visible, but*

*given the position of the TM5 helix, it does not appear to be that different from the IDR binding structure (5LZQ, 10A between two T211). The structures of in silico models are not shown in the figure, as it is only mentioned as modeled in Rossetafold. Please indicate their structures, especially focused on the relative orientation of T211 and S525 in the dimer, which would allow readers to determine the distances.*

We acknowledge the reviewer's concerns regarding Figure EV8 and the DEER data for T211R1. Upon re-evaluation, we recognize that the non-oscillating nature of the DEER data for T211R1 leads to broad distributions, indicating increased conformational dynamics, which is expected for a highly dynamic loop. Consequently, we have limited the discussion and interpretation of T211R1 in the revised manuscript and focused more on C599R1.

**Reviewer #3 (Recommendations for the authors):**

*A careful interpretation of the data in view of these limitations and without directly linking to asymmetry could solve the problem of the over-interpretation of the DEER data.*

We respectfully disagree with the reviewer. Please see our detailed response above.

*Additional comments:*

*(1) Did the authors use a Cys-less construct for spin labeling and DEER experiments?*

We utilized a nearly Cys-less construct in which all native cysteines were mutated to serine, except for Cys183, which was retained due to its buried location and functional importance. We then introduced single cysteine mutations for spin labelling. For C599, Ser599 was reverted to cysteine.

*(2) The time data for position T211R1 is too short for most cases (Figure EV8D) for a reliable distance determination. No confidence interval is given for the '+Ca' sample distance distributions.*

We recorded longer time traces for two of the conditions to better assign the background. We did not use the 211R1 data to reach any conclusions regarding asymmetry, which were based on the 525R1 and the 599R1 data. We now simply include T211R1 data to indicate the high mobility observed at loop5-6. We have added the confidence interval for the +Ca condition.

*(3) It is recommended to mention the 2+1 artefact obvious at the end of the DEER data.*

In the methods section, we have mentioned that the “2+1” artefact present at the end of the S525R1, and T211R1 DEER data likely arises from using a 65 MHz offset, rather than an 80 MHz offset (as for the C599R1 data), which avoids significant overlap of the pump and detection pulses. We also mention in the methods section that owing to the intense “2+1” artefact, the decision was made to truncate the artefact away, to minimise the impact on data treatment. As for motivation to use the lower offset of 65 MHz, we did so to maximise the achievable signal-to-noise ratio (SNR), as particularly for the T211R1 data, the detected echo was quite weak. This was further exacerbated by the poor transverse relaxation time observed at that site.

*(4) Please check the number of significant digits for all the reported values.*

We have addressed the number of significant digits as requested.



(5) Please report the mean distances from DEER experiments with the standard deviation or FWHM.

We have addressed this in the revised manuscript, we report modal distances rather than the mean distances and provide the FWHM and standard deviation.

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